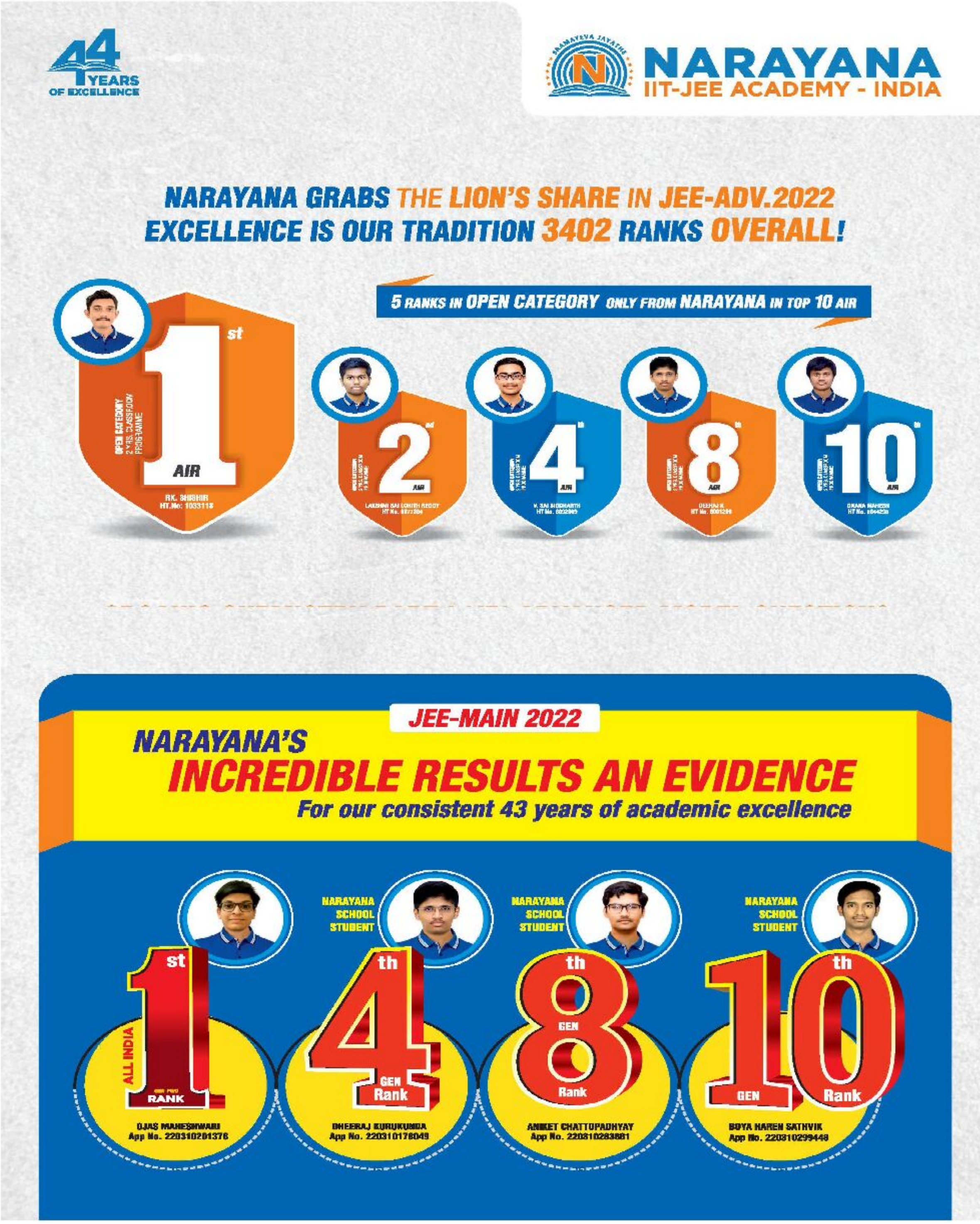


CHEMISTRY

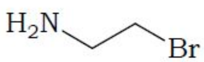
ORGANIC CHEMISTRY PART-II(SR)

JEE ADVANCED IMPORTANT QUESTIONS

ORGANIC CHEMISTRY PART-II(SR)

SINGLE ANSWER TYPE

1. The rate of cyclization of the following halo amines follows the order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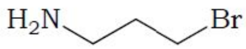
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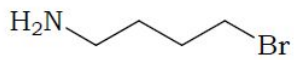
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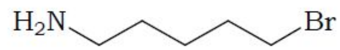
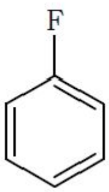

[S]

A) [R]>[S]>[P]>[Q] B) [S]>[R]>[P]>[Q] C) [P]>[Q]>[R]>[S] D) [R]>[S]>[Q]>[P]

2. Butane nitrile is best prepared by

A) Reaction of butan-1-ol with KCN
B) Reaction of but-1-ene with HCN
C) Reaction of 1-iodoobutane with KCN
D) Reaction of 1-chloropropane with KCN

3. Ease of aromatic nucleophilic substitution among these compounds will be in the order as: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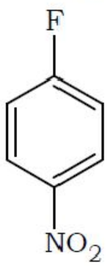
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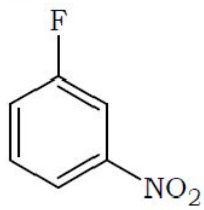
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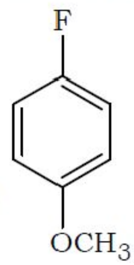
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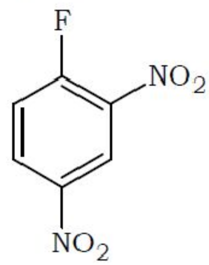
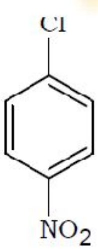
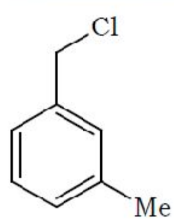
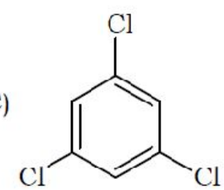
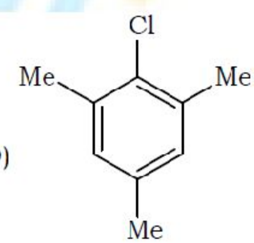
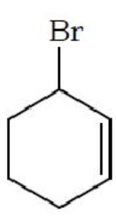
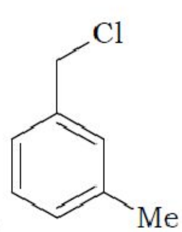

V

A) (V) > (II) > (III) > (I) > (IV)
B) (IV) > (I) > (III) > (II) > (V)
C) (IV) > (III) > (II) > (V) > (I)
D) (V) > (III) > (II) > (IV) > (I)

4. Which of the following compound undergoes hydrolysis most readily with aq. NaOH?

A) 
B) 
C) 
D) 

5. Which of the following compounds rarely undergo S_N2 reaction?


A)

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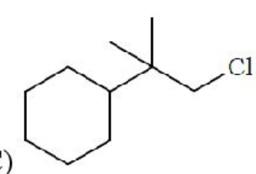
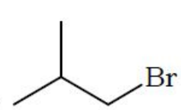
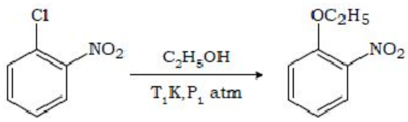
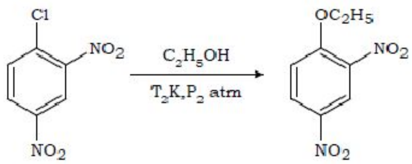
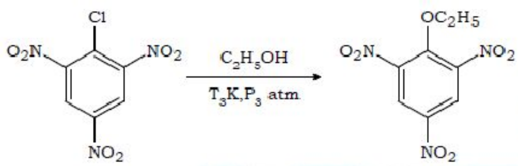

C)

D)

6. Let us consider the following reactions:

I


II


III


Where T_1, T_2, T_3 are temperatures while P_1, P_2, P_3 are pressures. Select the correct one.

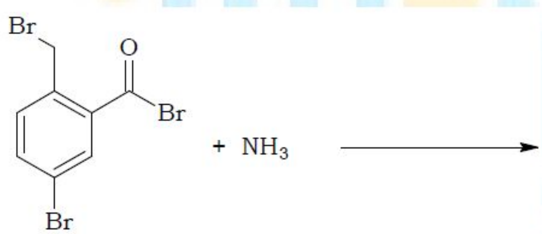
A) T_1 is the lowest temperature and P_3 is the highest pressure.

B) T_3 is the highest temperature and P_3 is the lowest pressure.

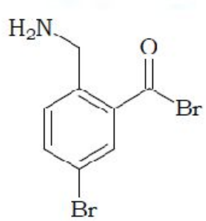
C) T_2 is the lowest temperature and P_2 is the lowest pressure.

D) T_3 is the lowest temperature and P_3 is the lowest pressure.

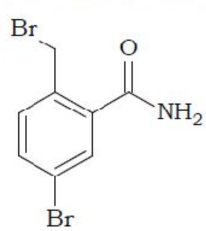
7. The product of the following reaction is



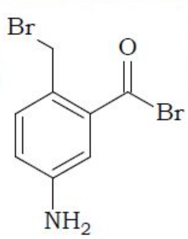
(1.0 mol) (1.0 mol)



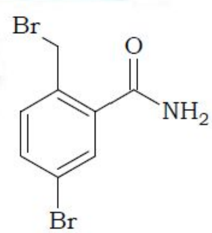
A) (1.0 mol)



B) (1.0 mol)



C) (1.0 mol)



D) (0.5 mol)

8. 8. The product formed when p-chlrorotoluene is treated with aq.NaOH at 340 °C and the mechanism by which it is formed is

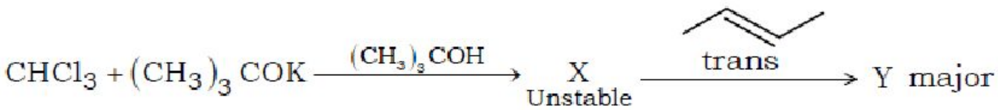
A) p-cresol, addition – elimination

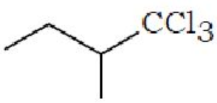
B) mixture of m-cresol and p-cresol, elimination – addition

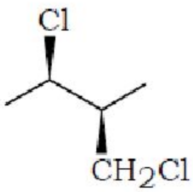
C) mixture of o-cresol and m-cresol, elimination – addition

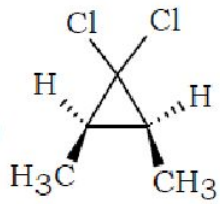
D) p-quinol, addition – elimination

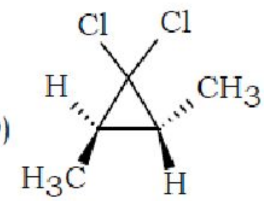
9. The end product in the following reaction is



A) 

B) 

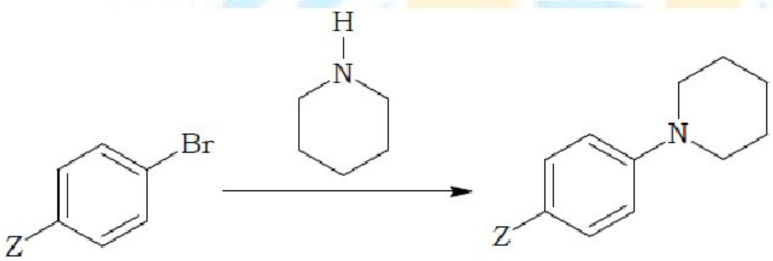
C) 

D) 

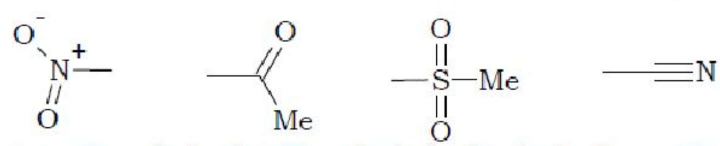
10. Transition state with least double bond character will be formed when methoxide reacts with

A) 2-iodohexane to form2-hexene
B) 2-iodohexane to form 1-hexene
C) 2-fluorohexane to form 2-hexene
D) 2-fluorohexane to form 1-hexene

11. Correct order of reactivity of different substituted bormobenzenes towards the following reaction is

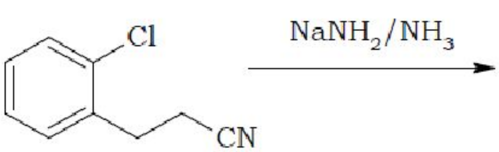


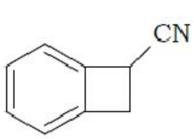
I II III IV

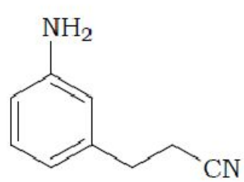
Z = 

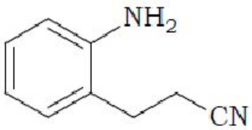
A) I>II>III>IV
B) IV>I>II>III
C) I>III>IV>II
D) III>I>IV>II

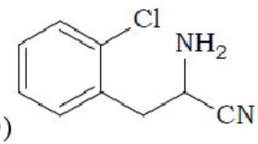
12. Product of the following reaction is



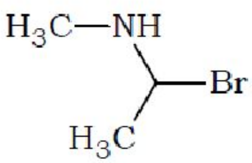
A) 

B) 

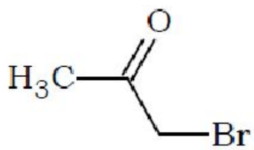
C) 

D) 

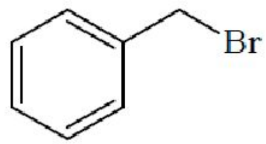
13. The correct order of hydrolysis (S_N1) of the following halides is



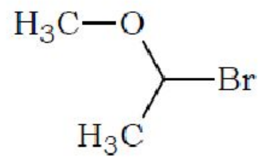
1



2



3



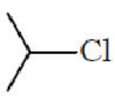
4

A) 2>1>3>4 B) 2>1>4>3 C) 4>1>3>2 D) 1>4>3>2

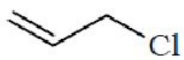
14. NaCN In DMF, undergoes S_N2 reaction with each of P, Q,R and S . The rates of the reaction vary as

H_3C-Cl

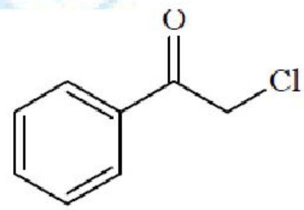
P



Q



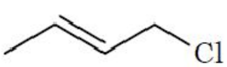
R



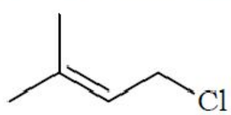
S

A) P > Q > R > S B) S > P > R > Q C) P > R > Q > S D) R > P > S > Q

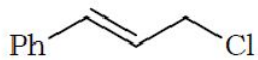
15. Correct order of reactivity towards S_N1 reaction of the following compounds is



I



II



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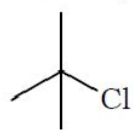
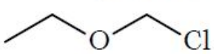


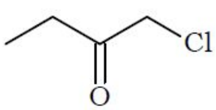
IV

A) II > III > I > IV B) II > III > IV > I C) III > II > I > IV D) III > II > IV > I

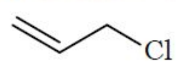
16. Correct order of reactivity towards S_N2 reaction of the following compounds is



I



II



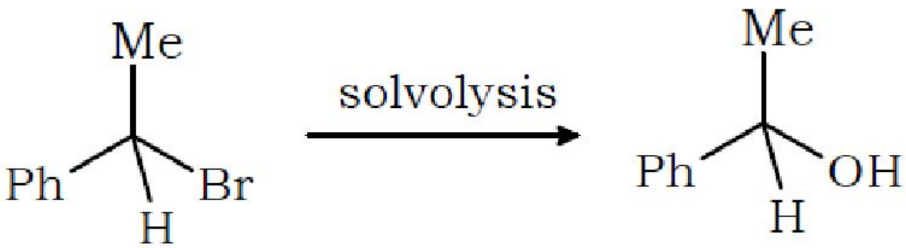
III

CH_3Cl

IV

A) II > I > IV > III B) II > I > III > IV C) I > II > III > IV D) I > II > IV > III

17. Consider the reaction

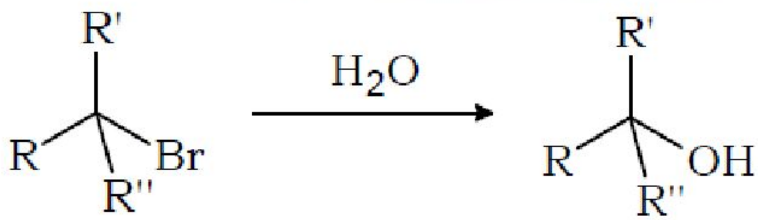


Which mixed solvent will give maximum racemization?

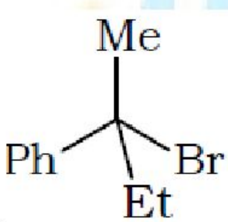
A) 20% acetone + 80% H_2O B) 40% acetone + 60% H_2O

C) 60% acetone + 40%H_2O D) 80% acetone + 20% H_2O

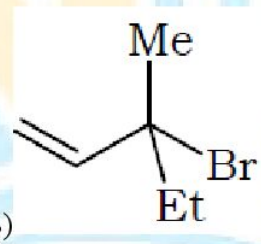
18. For the reaction



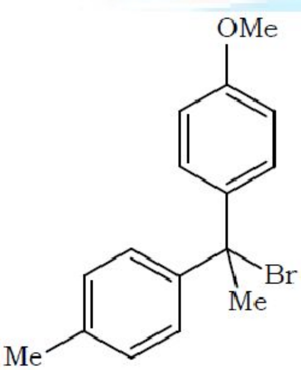
which of the following substrates will give maximum racemization?



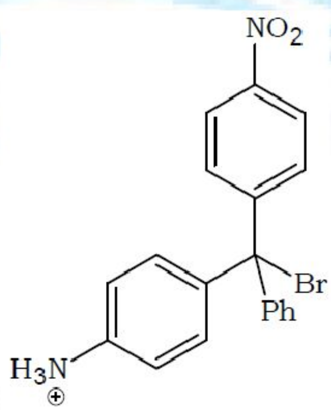
A)



B)



C)



D)

19. Which of the following undergoes maximum inversion of configuration with aq.NaOH?

A) MeI B) $MeCH_2I$ C) Me_2CHI D) Me_3CI

20. Isopropyl chloride undergoes SN1 reaction faster than S_N2 reaction with aqueous NaOH. Which of the following statement explains this observation?

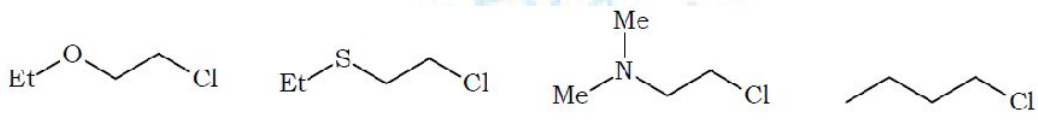
A) Carbocation intermediate is more stable than the transition state of S_N2.

B) S_N1 reaction has a positive entropy of activation $\left(\Delta S^{\ddagger}\right)$ while S_N2 has a negative entropy of activation.

C) Reaction is exothermic by S_N1 while endothermic by S_N2.

D) Water acts as the nucleophile in S_N1 while hydroxide ion acts as the nucleophile in S_N2

21. The correct order of reactivity of the following compounds towards aq.NaOH is



I II III IV

A) II > III > IV > I B) II > III > I > IV C) I > II > III > IV D) III > II > I > IV

22. Addition of small amounts of NaI catalyses the reaction $RCl + RONa \rightarrow ROR + NaCl$. Which of the following statement(s) explain(s) this observation correctly?

A) I^- is a better nucleophile as well as a better leaving group than Cl^-

B) I^- is a stronger base than Cl^-

C) I^- is a weaker nuclephile than Cl^-

D) I^- is a weaker base than Cl^-

23. In the following two phase reaction, catalyst works by

$$C_6H_5CH_2Br + KCN \xrightarrow[H_2O/\text{heptane}]{C_6H_5CH_2\overset{+}{N}(CH_3)_3\bar{C}l} C_6H_5CH_2CN + KBr$$

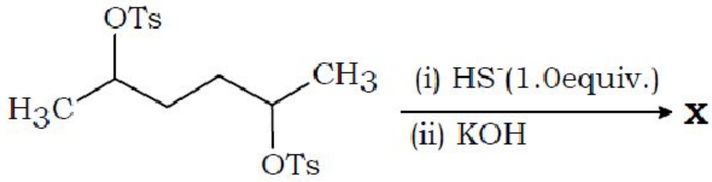
A) removing K^+ from the organic phase to make CN^- more nucleophilic

B) transporting $C6H_5CH_2Br$ from organic phase to the aqueous phase containing CN^-

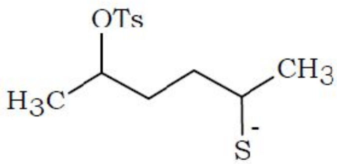
C) removing Br^- from the organic base to aqueous phase

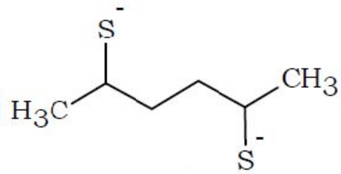
D) transporting CN^- from the aqueous phase to organic phase containing $C_6H_5CH_2Br$

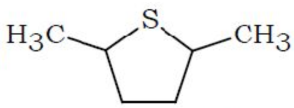
24. In the give reaction

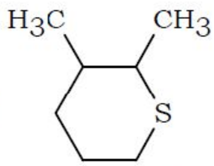


X is

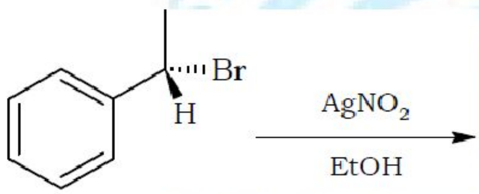
A) 

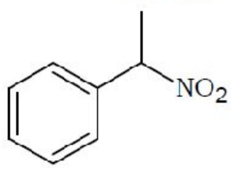
B) 

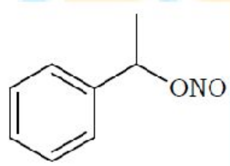
C) 

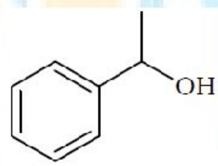
D) 

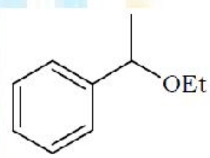
25. Major product of the following reaction is



A)  *dl*-pair

B)  *dl*-pair

C)  *dl*-pair

D)  *dl* pair

26. Rates of the following nucleophiles towards the reaction with MeBr in ethanol follow the order

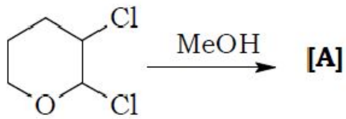
A) $Et_3N < PhO^- < EtO^- < PhS^-$

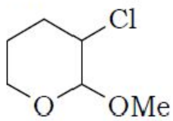
B) $PhS^- < EtO^- < PhO^- < Et_3N$

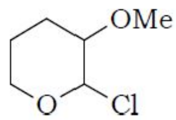
C) $PhO^- < Et_3N < PhS^- < EtO^-$

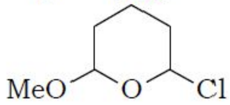
D) $PhO^- < EtO^- < Et_3N < PhS^-$

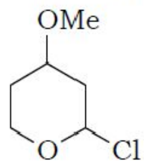
27. Compound [A] in the following reaction is



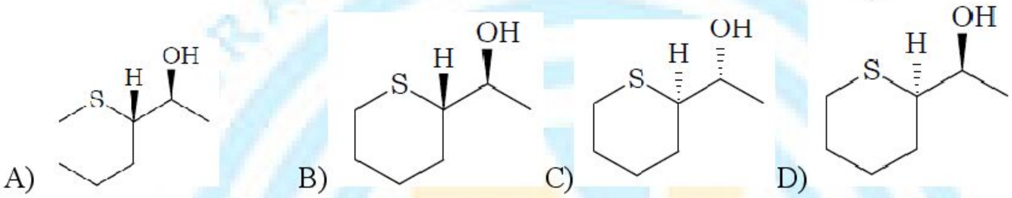
A) 

B) 

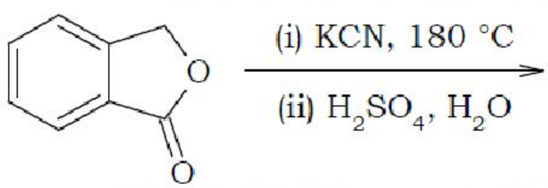
C) 

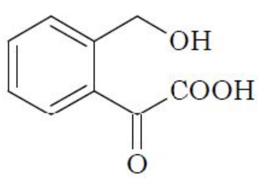
D) 

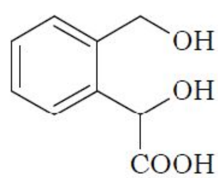
28. Product of the following reaction with stereochemistry is

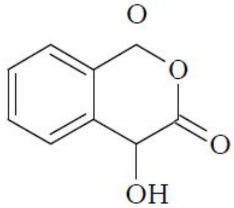


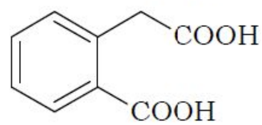
29. Compound [F] in the following reaction is



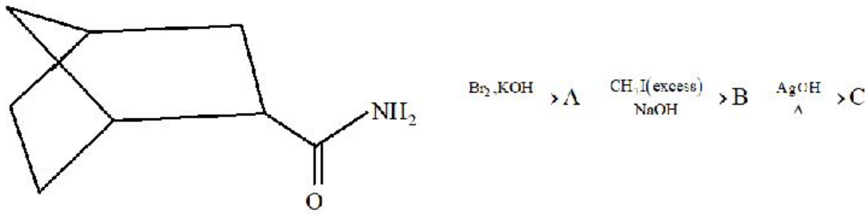
A) 

B) 

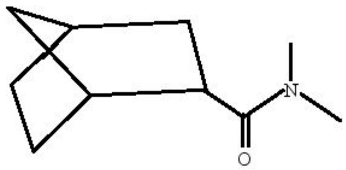
C) 

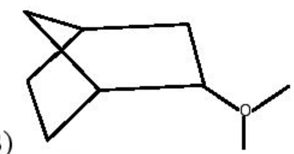
D) 

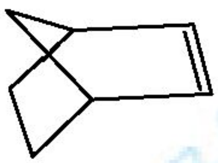
30. Consider the following sequence of reaction



Identify product C

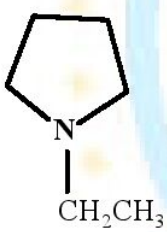
A) 

B) 

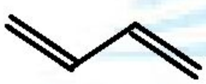
C) 

D) 

31. What would be the final product of reaction

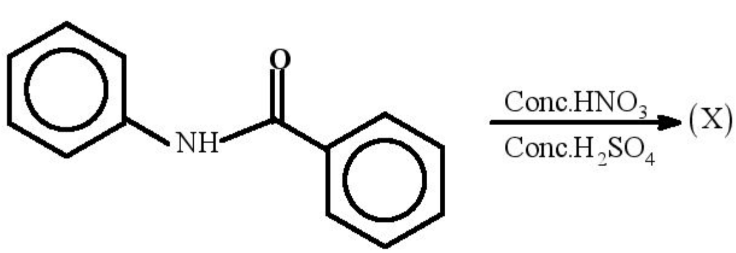
A) 

B) 

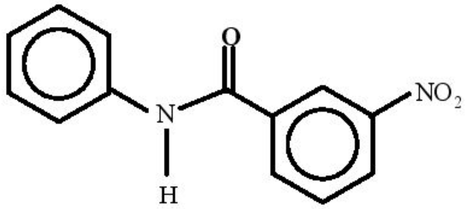
C) 

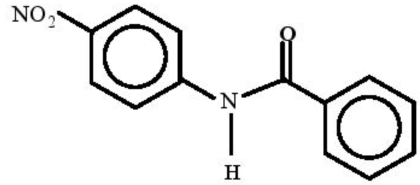
D) 

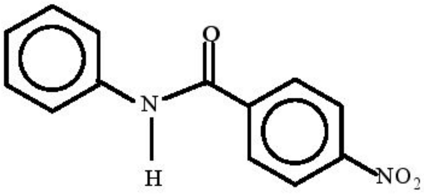
32. In the following reaction

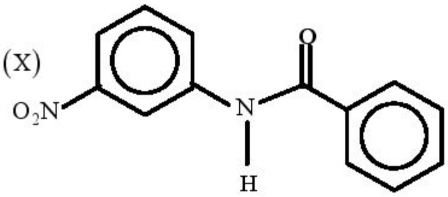


the structure of the major product (X) is

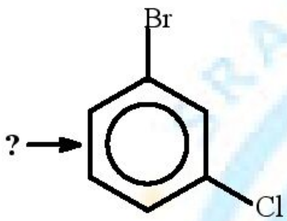
A) 

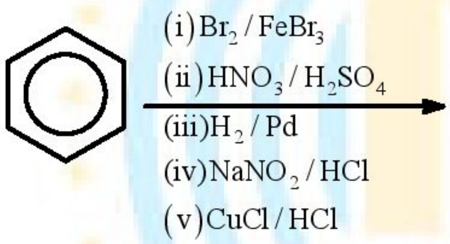
B) 

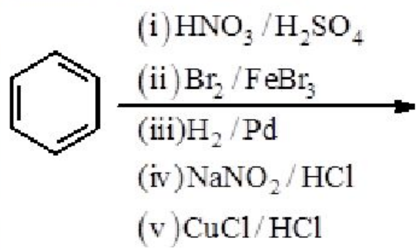
C) 

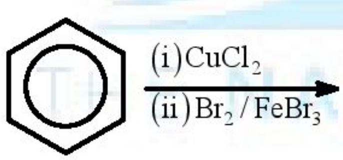
D) (X) 

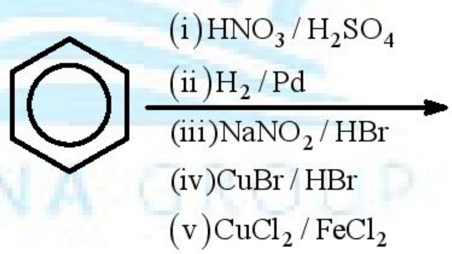
33. Choose the best method for the preparation of the following compound



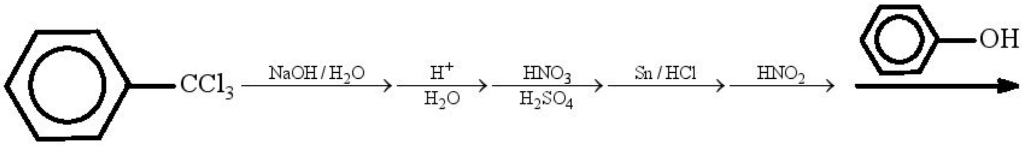
A) 

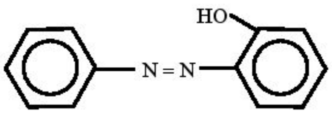
B) 

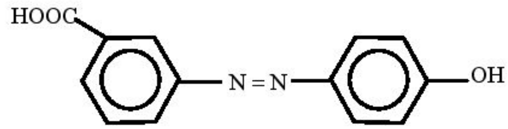
C) 

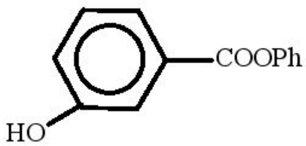
D) 

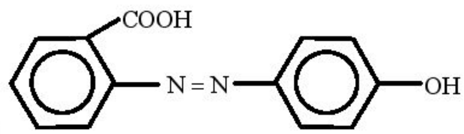
34. Select the final product of the following sequence of reaction

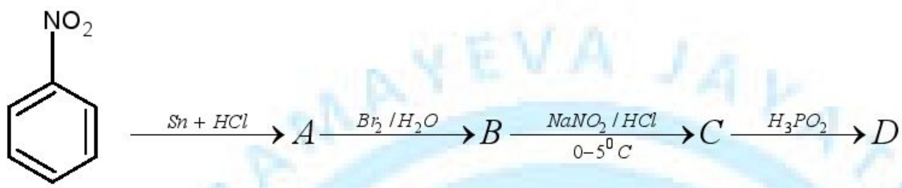


A) 

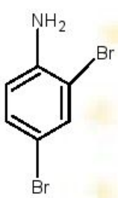
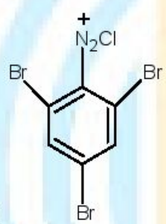
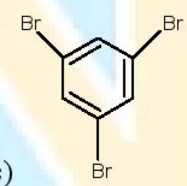
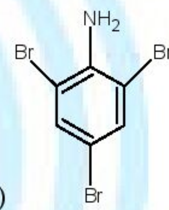
B) 

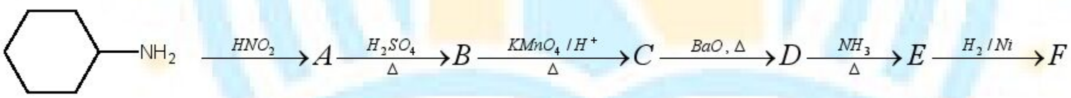
C) 

D) 

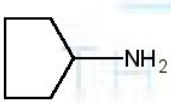
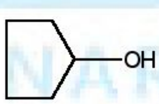
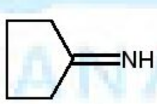
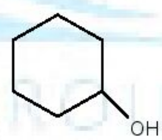
35. 

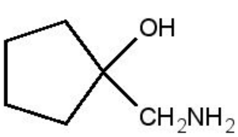
In the above sequence of reactions, final product D is

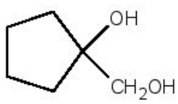
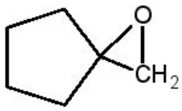
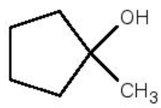
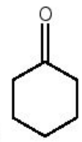
a) 
b) 
c) 
d) 

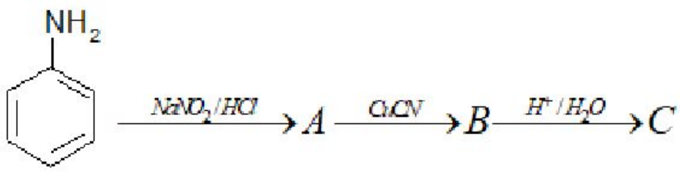
36. 

In the above sequence of reactions, final product F is

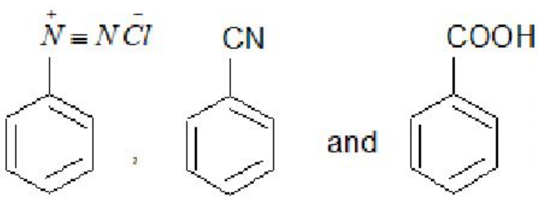
a) 
b) 
c) 
d) 

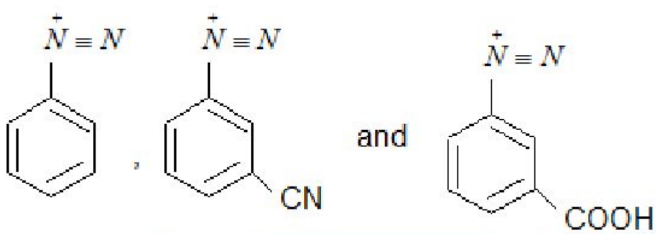
37.  $\xrightarrow{HNO_2} A$ (*major product*), *A is*

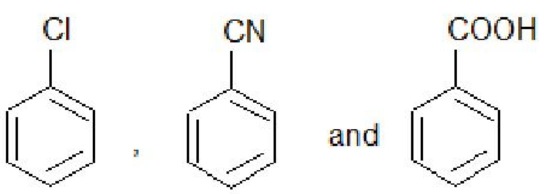
a)  b)  c)  d) 

38. 

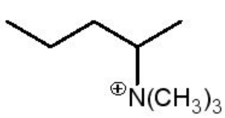
A, B and C are

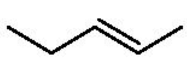
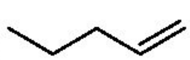
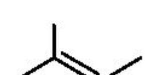
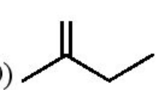
A) 

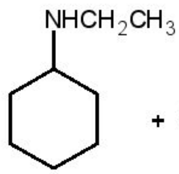
B) 

C) 

D) None

39.  $\xrightarrow[\Delta]{OH^-}$ major product (X). X is

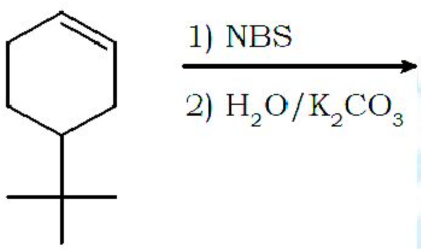
A)  B)  C)  D) 

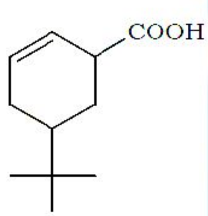
40.  + CH_3I (excess) → product. The product is

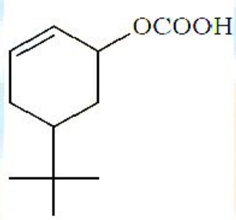
A) 1^o – amine B) 2^o – amine

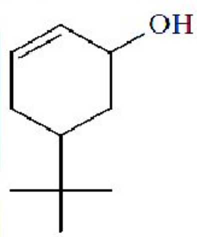
C) 3^o – amine D) quaternary ammonium salt

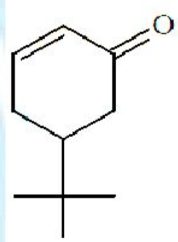
41. The product of the reaction given below is



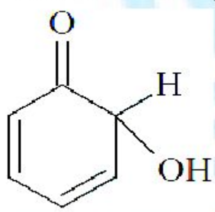
A) 

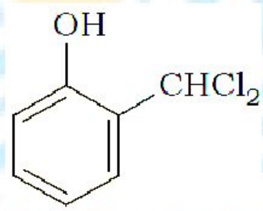
B) 

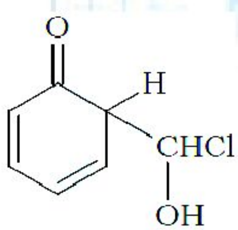
C) 

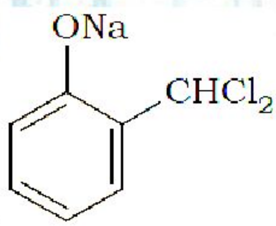
D) 

42. When phenol is reacted with $CHCl_3$ and NaOH followed by acidification, salicyladehyde is obtained. Which of the following species is the intermediate in the reaction

A) 

B) 

C) 

D) 

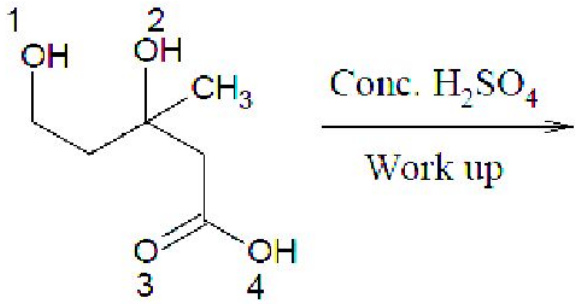
43. Which carboxylic acid does not give racemic product upon reaction with bromine in presence of red phosphorus?

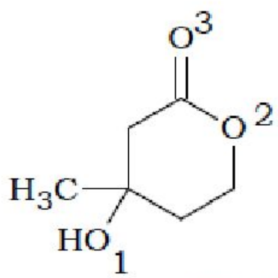
Note : Consider all carboxylic acids are optically pure

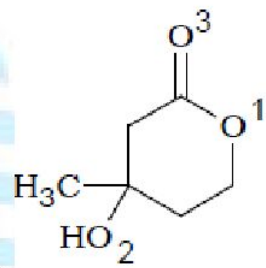
A) 2-methyl butanoic acid B) 2-methyl pentanoic acid

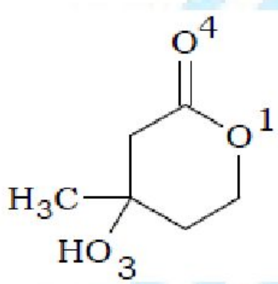
C) 2,3-dimethyl butanoic acid D) 3-methyl pentanoic acid

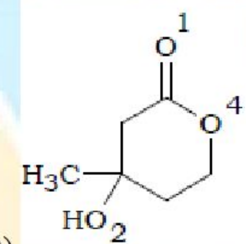
44. The product of the given reaction is



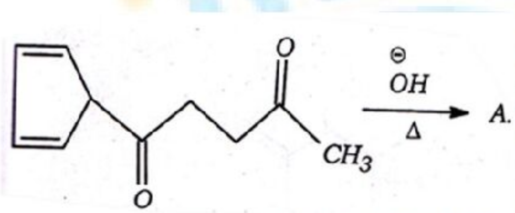
A) 

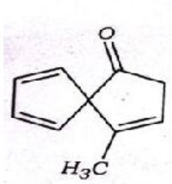
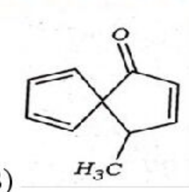
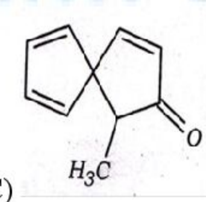
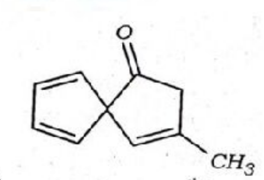
B) 

C) 

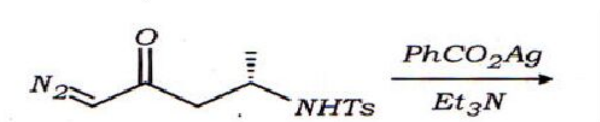
D) 

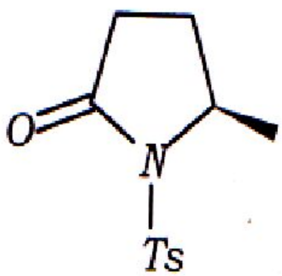
45. What is the product A?

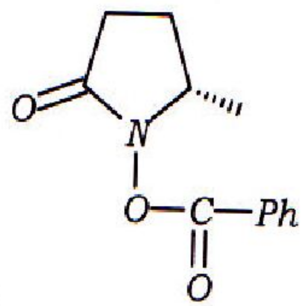


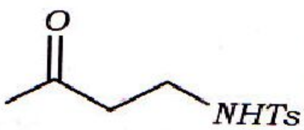
A) 
B) 
C) 
D) 

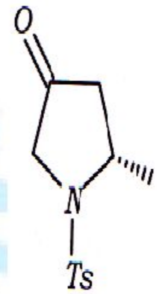
46. Predict the product of the diazoketone with silver (I) benzoate



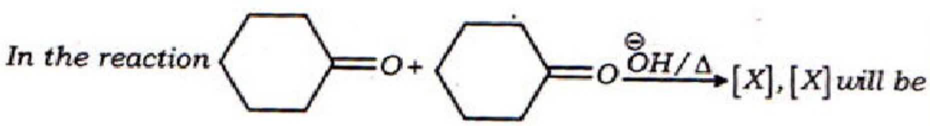
A) 

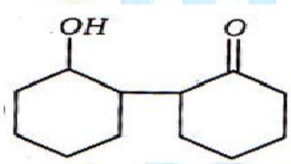
B) 

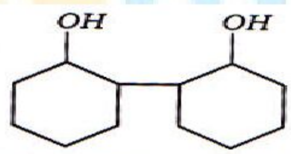
C) 

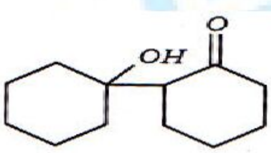
D) 

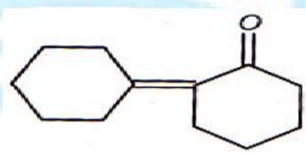
47.



A) 

B) 

C) 

D) 

48. An ester (A) with molecular formula $C_9H_{10}O_2$ was treated with excess of CH_3MgBr and the complex so formed was treated with H_2SO_4 to give an olefin (B) as minor product. Ozonolysis of (B) gave a ketone with molecular formula C_4H_8O which shows positive iodoform test. The structure of (A) is

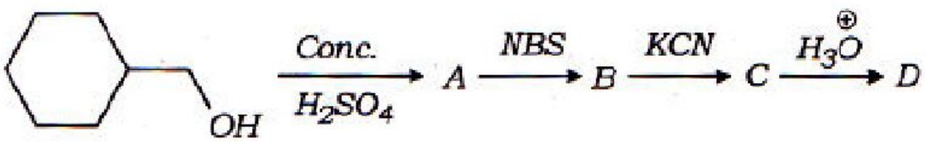
A) $C_6H_5COOC_2H_5$

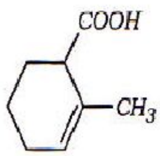
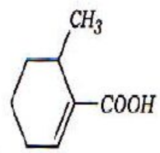
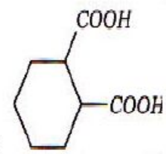
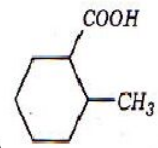
B) $H_4C_2-\overset{\overset{O}{\|}}{C}-O-C_6H_5$

C) $C_4H_8COOC_4H_9$

D) $H_{11}C_5-\overset{\overset{O}{\|}}{C}-O-C_3H_7$

49. Find out 'D' in the following sequence



A)  B)  C)  D) 

50. Compare the two methods shown for the preparation of carboxylic acids.

Method 1: $RBr \xrightarrow[\text{diethyl ether}]{Mg} RMgBr \xrightarrow[ii)H_3\overset{\oplus}{O}]{i)CO_2} RCO_2H$

Method 2: $RBr \xrightarrow{NaCN} RCN \xrightarrow[\Delta]{H_2OL,HCl} RCO_2H$

Which of the following statements correctly describes this conversion?

A) Both method 1 and method 2 are appropriate for carrying out this conversion

B) Neither method 1 nor method 2 is appropriate for carrying out this conversion

C) Method 1 will work well, but method 2 is not appropriate

D) Method 2 will work well, but method 1 is not appropriate

51. Which functional group participates in disulphide bond formation in proteins

A) Thioacetone B) Thiol C) Thioether d) Thioester

52. Which of the following is incorrect statement

A) Both glucose and fructose undergo mutarotation is aqueous solution.

B) Both glucose and fructose on treatment with $CH_3OH\,/\,dryHCl$ gas do not form α and β - methyl glucosides and fructosides.

C) Six membered cyclic structure of hexose is called pyranose form

D) trioses and tertroses have open chain structures

53. Antiseptics and disinfectants either kill (or) prevent growth of micro organisms. The incorrect statement is

A) Dilute solutions of boric acid and H_2O_2 are strong antiseptics

B) Disinfectants harm the living tissues

C) A 0.2 % solution of phenol is an antiseptic and 1% solution acts as a disinfectant

D) Chlorine and Iodine are used as strong disinfectants.

54. Arrange the following in increasing order of their inter molecular forces.

(I) Nylon-6,6 (II) Buna-S (III) Plythene

A) II, III, I b) III, II, I C) I, II, III D) II, I, III

55. Polymer with low degree of polymerisation is known as

A) higher polymer B) Oligomer C) Macromolecule D) Co-polymer

56. Which of the following amino acid has phenol group

A) Tryptophan B) Proline C) Histidine D) Tyrosine

57. Order of decreasing inter molecular forces in

(I) Nylon 6 (II) Neoprene (III) Poly Vinyl chloride

A) $I > II > III$ B) $III > II > I$ C) $I > III > II$ D) $II > III > I$

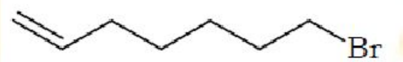
MORE THAN ONE ANSWER TYPE

58. Which of the following do not give the expected Grignard reagent on reaction with magnesium in dry ether?

A) 

B) 

C) 

D) 

59. Which of the following is/are the correct order of nucleophilicity in a weakly polar aprotic solvent like acetone?

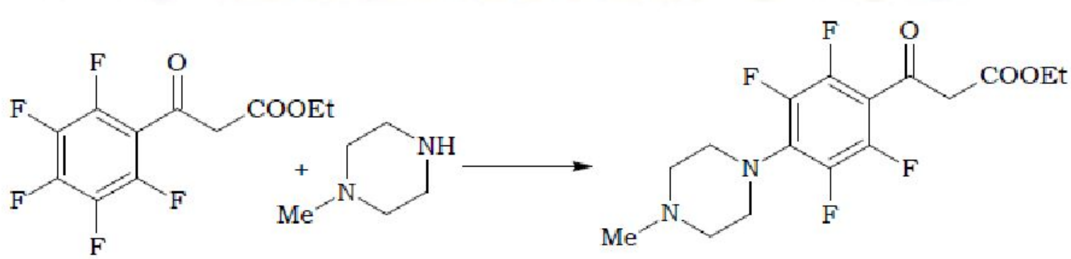
A) LiI > LiBr > LiCl > LiF

B) CsF > RbF > KF > NaF > LiF

C) Bu_4N+Cl- > Bu_4N+Br- > Bu4N+I^-

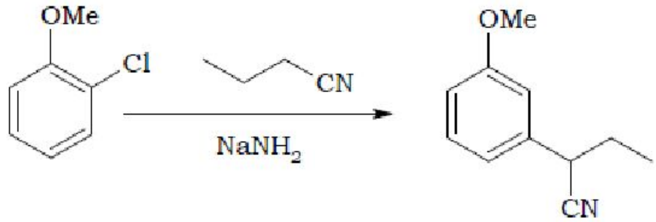
D) $CH_3Na > NaNH_2 > NaOH > NaF$

60. Correct reaction(s) among the following is/are

A) 

B) 

C) 

D) 

61. Consider the following schemes:

Scheme 1: Ph_3CCl on reaction with metallic silver gives a dimer, which dissolves in benzene to give a yellow solution. Addition of I_2 renders it colourless.

Scheme 2: Ph_3COH on reaction with HBF_4 in acetic anhydride solvent gives a bright orange coloured crystalline salt precipitated from anhydrous ether. When water is added, the colour is discharged.

Scheme 3: An ether solution of Ph_3CCl, stirred with a strong reducing agent like Na/Hg, develops a deep red colour that is discharge almost immediately when water or ethanol is added.

The correct schemes and/or reason for the colour is/are

A) **Scheme 1**

$$Ph_3CCl + Ag \longrightarrow \underset{\text{(yellow)}}{Ph_3CCPh_3} \xrightarrow{I_2} \underset{\text{(Colourless)}}{Ph_3CI}$$

B) **Scheme 2**

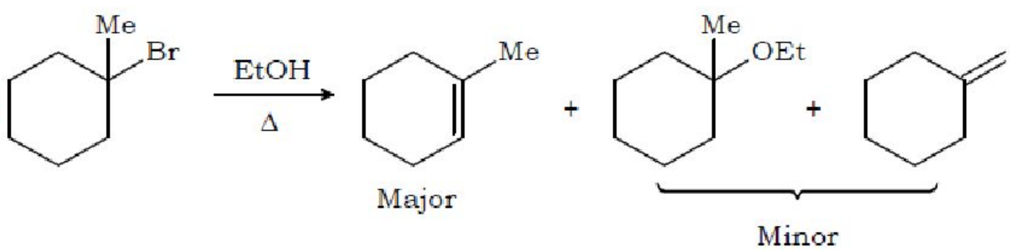
$$Ph_3COH + HBF_4 \longrightarrow \underset{\text{(Orange)}}{Ph_3\overset{+}{C}\,BF_4^-} \xrightarrow{H_2O} \underset{\text{(Colourless)}}{Ph_3COH}$$

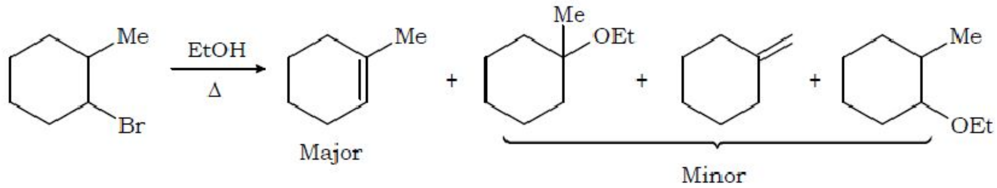
C) **Scheme 3**

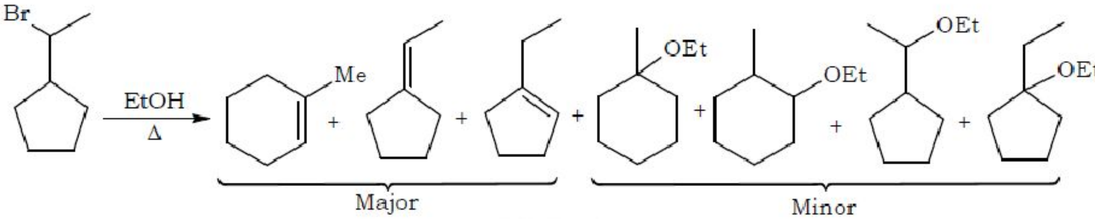
$$Ph_3CCl + Na(Hg) \longrightarrow \underset{\text{(Red)}}{Ph_3\overset{-}{C}\,Na^+} \xrightarrow{H_2O} \underset{\text{(Colourless)}}{Ph_3CH}$$

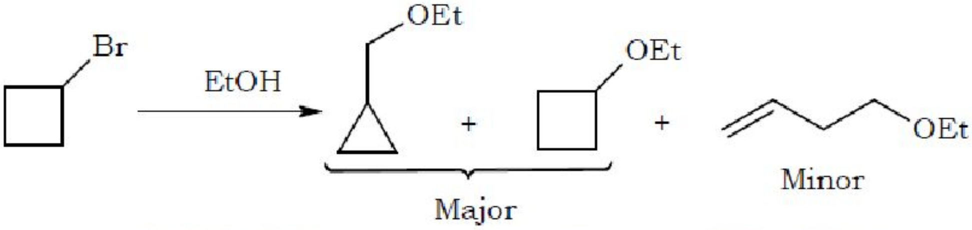
D) Colour in all the three schemes is attributed to extensive conjugation in the product formed.

62. Correct reaction(s) among the following is/are

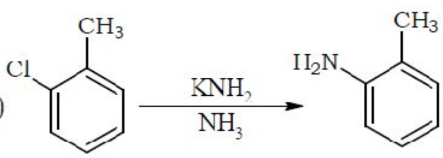
A) 

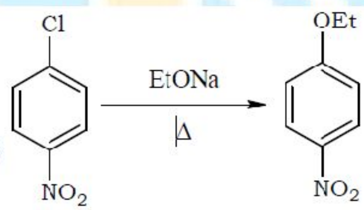
B) 

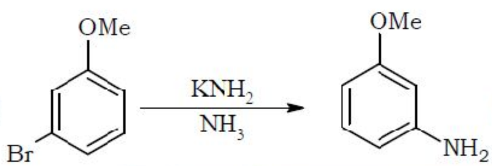
C) 

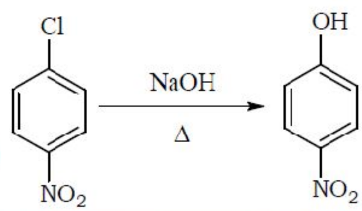
D) 

63. In which of the following reactions, the given product is the major product?

A) 

B) 

C) 

D) 

64. Biphenyl can be prepared by

A) Treating chlorobenzene with Na/dry ether

B) Heating iodobenzene with metallic copper

C) Treating benzene with chlorobenzene in presence of $AlCl_3$

D) Treatment of benzene with cyclohexene and HF at 273 K followed by treatment with Pt.

65. In Dow's process for the manufacture of phenol, PhCl is fused with NaOH at elevated temperature under pressure.

$$PhCl \xrightarrow[623K,300atm]{Naoh} [\text{Intermediate}] \xrightarrow{H_2O} \text{Phenol} + \underbrace{B+C}_{\text{Side product}}$$

Which of the following statements are correct?

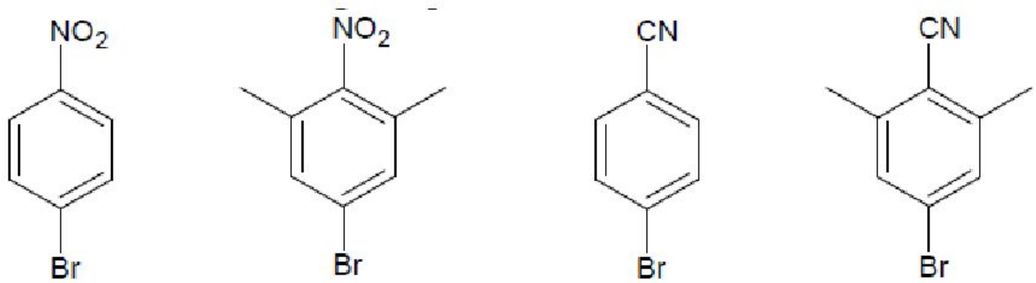
A) Phenol is formed via the formation benzyne intermediate

B) p-phenylphenol is also formed as a by-product

C) Diphenylether is also formed as a by-product

D) Biphenyl is also formed as a by-product

66. Consider the following compounds:



I II III IV

Choose the correct statement(s) from amongst the following regarding the reactivity of these compounds towards nucleophilic substitution:

A) I is more reactive than III.

B) The difference in reactivities between I and II is more than that between III and IV.

C) II is less reactive than I while IV is more reactive than III.

D) IV is most likely to react at a slower rate than II.

67. Vinyl chloride is less reactive than ethyl chloride towards nucleophilic substitution reactions because

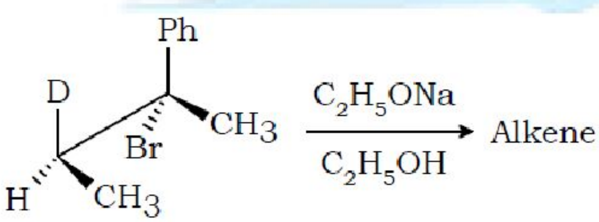
A) C – Cl bond in vinyl chloride is stronger due to higher percentage ‘s’ character of vinyl carbon

B) resonance increases the bond strength of C – Cl bond in vinyl chloride

C) the electrons of π bonds repel the nucleophile

D) there is partial double bond between carbon and chlorine in vinyl chloride

68. Consider the following reaction,



I.

The correct statement concerning the above reaction is

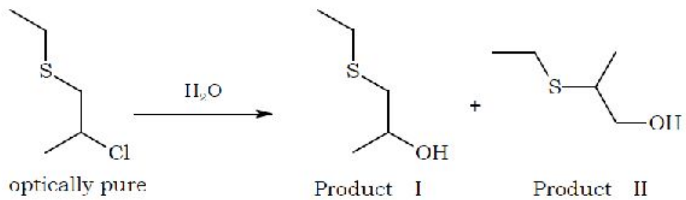
A) DBr is the eliminated and trans isomer is formed as the major product

B) HBr is eliminated and trans isomer is formed as the major product

C) diastereomer of (I) gives cis alkene as the major product by elimination of HBr

D) diastereomer of (I) give cis alkene as the major product by elimination of DBr

69. Consider the following solvolysis reaction



Correct statement(s) regarding this reaction is/are

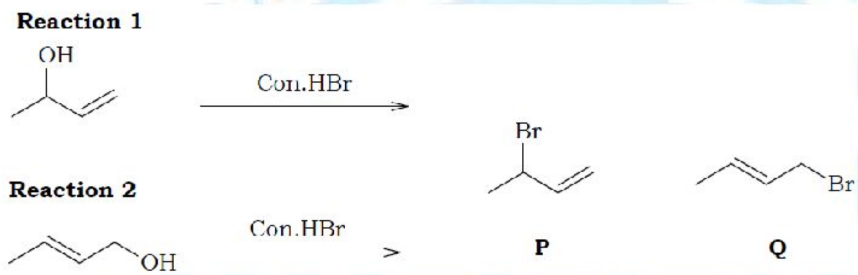
A) Product I is formed as a major

B) Product II is formed as a major

C) Product I is obtained with 100% retention of configuration

D) Product II is obtained with 100% inversion of configuration

70. Correct statement about the product(s) of the following reaction is



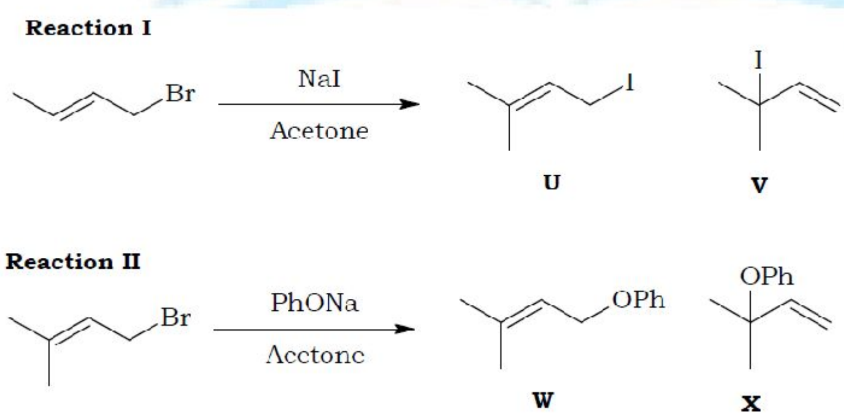
A) P is major product in Reaction 1 while Q is major product in Reaction 2

B) Q is major product in Reaction 1 while P is major product in Reaction 2

C) Q is major product and P is minor product in both Reaction 1 as well as Reaction 2.

D) Both the reactions follow S_N1 mechanism.

71. Correct statement about the product(s) of the following reaction is



A) U is major product in Reaction I and the mechanism is S_N2

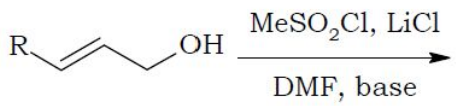
B) W is major product in Reaction II and the mechanism is S_N2

C) V is major product Reaction I and the mechanism is S_N2'

D) X is major product Reaction II and the mechanism is S_N2'

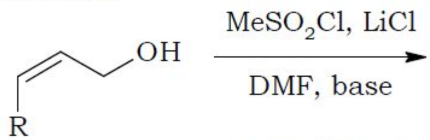
72. Correct statement(s) regarding the following reactions is/are

Reaction I



P **Q**

Reaction II



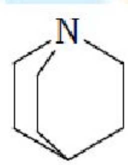
A) P is the product in both Reaction I and Reaction II

B) Q is the product in both Reaction I and Reaction II

C) SN2 is the mechanism followed in both Reaction I and Reaction II

D) SN1 is the mechanism followed in both Reaction I and Reaction II

73. The rate of reaction of methyl iodide with quinuclidine was measured in nitrobenzene, and then the rate of the reaction of methyl iodide with triethylamine was measured in the same solvent. Similarly the experiment was carried out replacing methyl iodide with isopropyl iodide. The concentration of the reagents was the same in all experiments. Correct statement(s) among the following regarding these reactions is/are



Quinuclidine

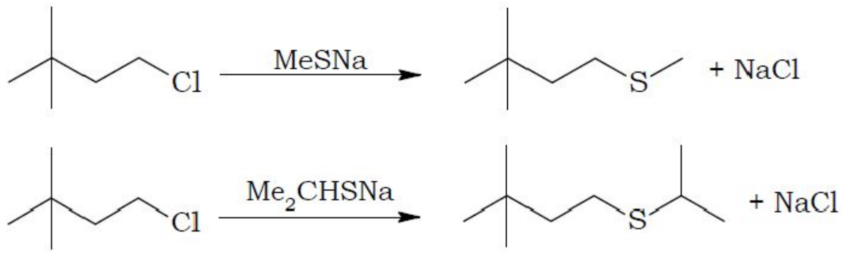
A) Rate constant for the reaction of methyl iodide with quinuclidine is greater than that of triethylamine.

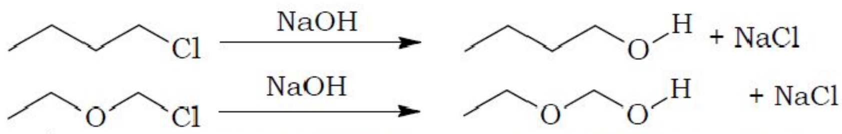
B) Triethlyamine is more nucleophilic than quinuclidine while quinuclidine is more basic than triethylamine.

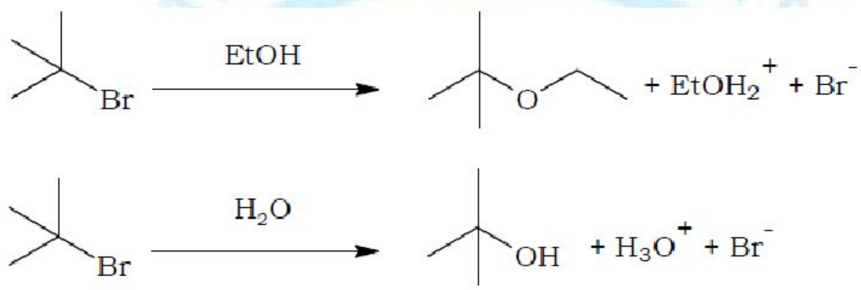
C) Rate of reaction with isopropyl iodide is less than that with methyl iodide for both nucleophiles.

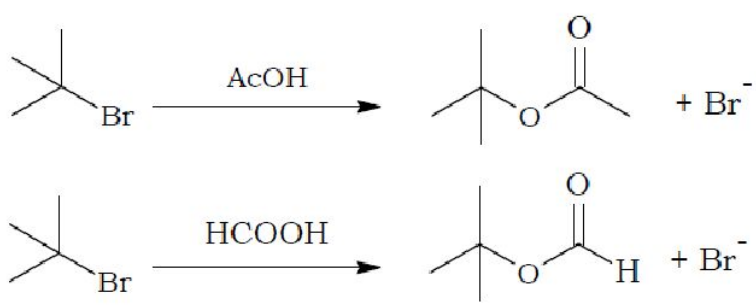
D) Lone pair of quinuclidine is more directional than that of triethylamine.

74. Consider the following pairs of reactions. In which of them, second reaction is faster than the first one?

A) 

B) 

C) 

D) 

75. Which of the following reagents give yellow precipitate with the compound $PhCOCH_2I$?

A) Ag_2O B) $I_2/NaOH$ C) I_2/Na_2CO_3 D) $KI/CuSO_4$

76. Which of the following sequence of reagent is the good means to furnish the conversion?

$R-CH_2OH \rightarrow R-CH_2NH_2$

A) $KMnO_4, SOCl_2, NH_3, \Delta, NaOBr$ B) $SOCl_2, NaCN, H_2/Ni$

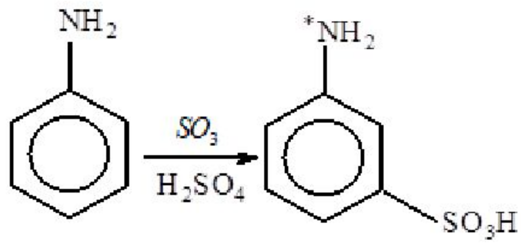
C) CrO_3 in dilute acetone, NH_3, H_2, Ni D) Cu, 300^0C, $NH_3, LiAlH_4$.

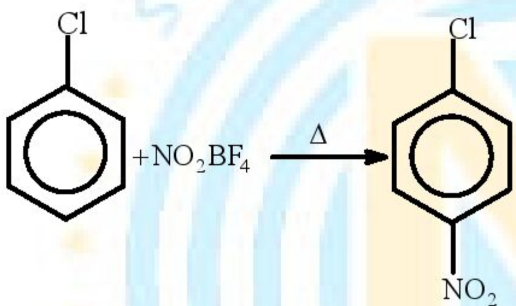
77. $$Ph-\overset{\overset{\large O}{\|}}{C}-NH_2 + H_3C-\overset{\overset{\large O}{\|}}{C}-\overset{15}{N}H_2 \xrightarrow{OH^- + Br_2} A + B$$

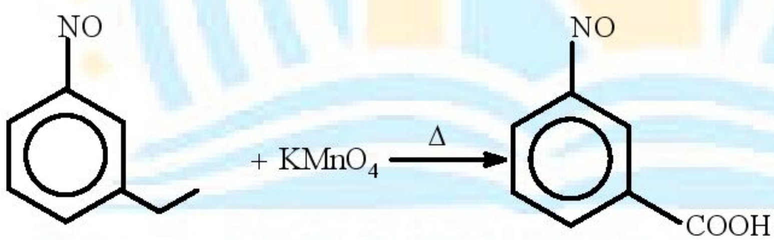
Product A and B are

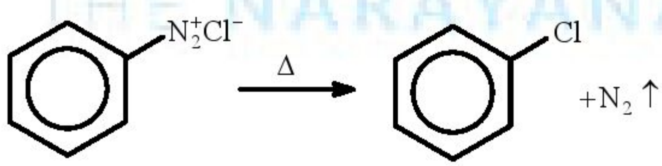
A) $Ph-NH_2$ B) CH_3-NH_2 C) $Ph-CH_2-NH_2$ D) $Ph-\overset{15}{N}H_2$

78. In which of the following, reactant and product are correctly matched?


A)


B)


C)


D)

79. $$\underset{(1^0\ amine)}{A} \xrightarrow[Excess]{CH_3I} B \xrightarrow[H_2O]{Ag_2O} C \xrightarrow{\Delta} D + one\ mole\ of\ H_2C=CH_2 + H_2O$$

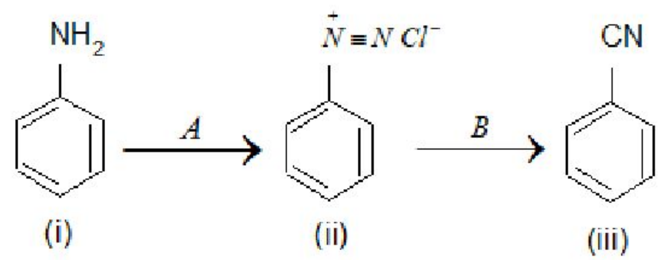
A) The amine A is Ethanamine B) 'D' is trimethyl amine

C) 'D' is triethyl amine D) 'C' is quaternary ammonium iodide.

80. Which of the following compound (s) is (are) more basic than $C_6H_5NH_2$

A) paranitro aniline
B) $C_6H_5NHCH_3$
C) paramethoxy aniline
D) orthomethoxy aniline

81.



Choose the correct statement(s) for the above reaction

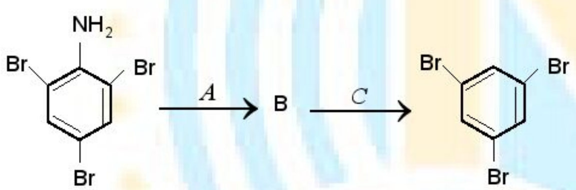
A) A behaves as an electrophile to attack benzene ring at o/p position.

B) I to II conversion is highly exothermic

C) B is CuCN

D) NH_2 group is a ring activating group for aromatic electrophilic substitution reaction for benzene.

82. Choose the correct statement (s) for the following conversion.



A) A is Electrophile

B) C is an monobasic acid

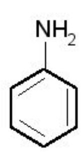
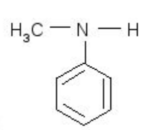
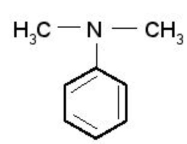
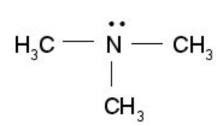
C) one of the biproduct is gas having highest bond order

D) conversion of reactant to B take place at high temperature

83. (S) – 2- methylbutanamide $\xrightarrow[OH^-]{Br_2}$ A $\longrightarrow$ B $\xrightarrow{H_2O}$ C (optically active amine)

A) A is a nitrene
B) B is an isocyanate
C) C is (R) - sec-butylamine
D) C is (S) – sec-butylamine

84. Which of the following shows carbyl amine reaction?

A)  B)  C)  D) 

85. $H_3C-CH_2-Cl \xrightarrow{NaCN} X \xrightarrow{Ni/H_2} Y \xrightarrow{Ac_2O} Z$. Z in this reaction

A) $CH_3-CH_2-CH_2-\overset{O}{\overset{||}{C}}-NH-CH_3$

B) $CH_3-CH_2-CH_2-NH_2$

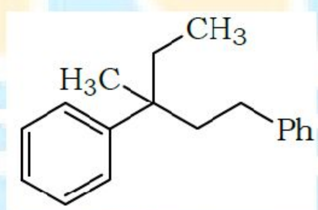
C) $CH_3-CH_2-CH_2-NH-\overset{O}{\overset{||}{C}}-CH_3$

D) $CH_3-CH_2-CH_2-\underset{O}{\underset{||}{C}}-NH-\underset{O}{\underset{||}{C}}-CH_3$

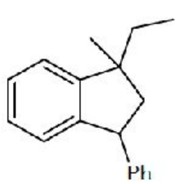
86. Secondary amine forms yellow oily liquid with HNO_2, which on warming with phenol and conc.H_2SO_4 gives a brown or red colour and which at once changes into blue-green. This reaction is called

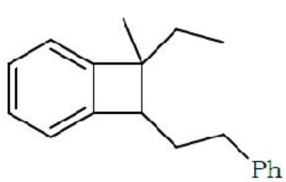
A) Carbyl amine reaction
B) Gabriel pthalamide reaction
C) Libermann's nitroso reaction
D) Hoffmann mustard oil reaction

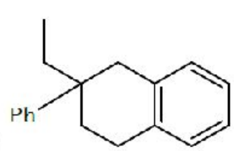
87. The following compound on monobromination with Br_2 in presence of light gives [X] as the major product along with low percentages of [Y] and [Z] and traces of [P] and [Q].

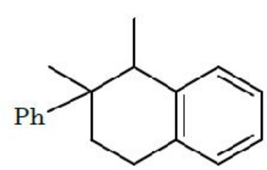


A mixture of [X], [Y] and [Z] is treated with anhydrous $AlBr_3$ in nitrobenzene as the solvent. Which of the following compound(s) may be found in the final product mixture?

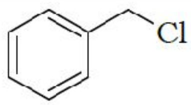
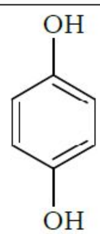
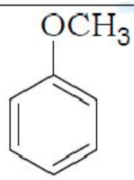
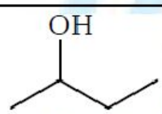
A) 

B) 

C) 

D) 

88. The structure(s) matched correctly with its/their names is/are

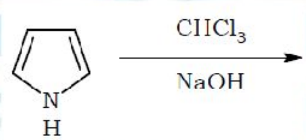
	Structure	Common Name	IUPAC name
A)		Benzyl chloride	Chlorophenylmethane
B)		Hydroquinone	Benzene-1,4-diol
C)		Anisole	Anisole
D)		sec-Butyl alcohol	Butan-2-ol

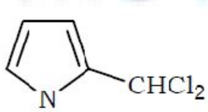
89. An organic compound $C_4H_8O_2$ reacted with excess $MeMgBr$ to give two isomeric compounds as products, of which at least one gives yellow precipitate with I_2 / Na_2CO_3. The compound may be

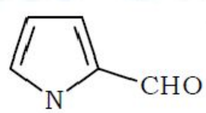
A) Propylmethanoate B) isopropylmethanoate

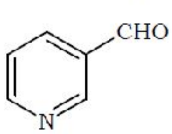
C) Ethylethanoate D) methylpropanoate

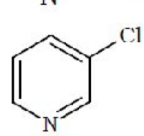
90. Product(s) of the following reaction is(are)



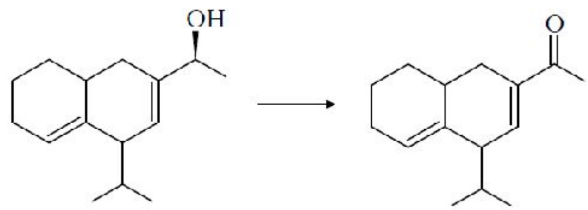
A) 

B) 

C) 

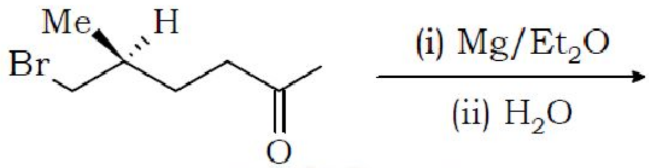
D) 

91. The following conversion can be effected using



A) NaOBr (1 equiv) B) Ceric ammonium nitrate C) Cu/573 K D) PCC

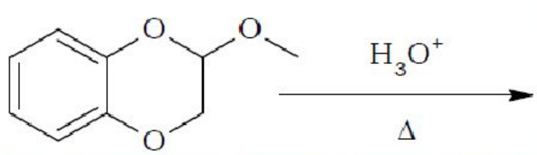
92. Correct statement(s) regarding the following reaction is/are



A) The product is an organometallic compound.
B) The product is a mixture of two isomeric compounds.
C) The product is an equimolar mixture of two diastereomers.
D) The product is a chiral ketone.

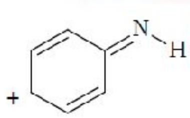
93. The white precipitate formed when phenol is treated with bromine water contains
A) 2-bromophenol B) 3-bromophenol C) 4-bromophenol D) 2,4,6-tribromophenol

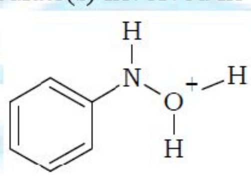
94. Correct statement(s) regarding the products of the following reaction is/are

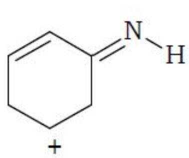


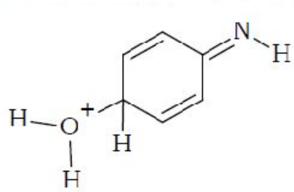
A) One of the products turns neutral ferric chloride violet.
B) One of the products gives Fehling's test.
C) All the products turn blue litmus red.
D) All the products evolve H2(g) on reaction with sodium metal.

95. Bamberger rearrangement involves conversion of N-Phenylhydroxylamine to aminophenols with aqueous sulphuric acid g. The possible intermediate(s) involved in this conversion is/are

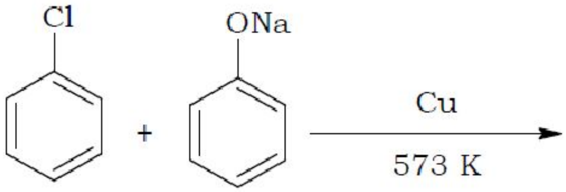
A) 

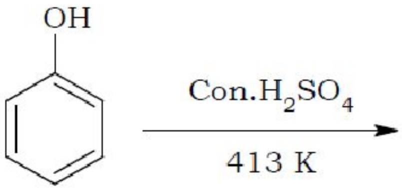
B) 

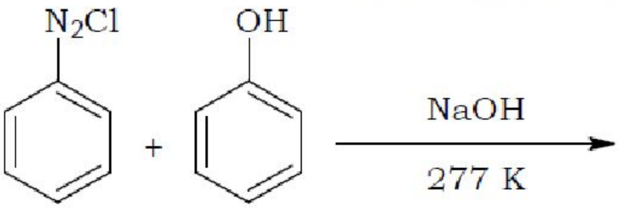
C) 

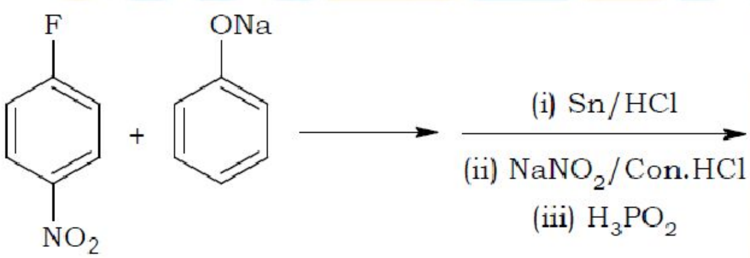
D) 

96. Which of the following methods may yield a diphenyl ether as the major product?

A) 

B) 

C) 

D) 

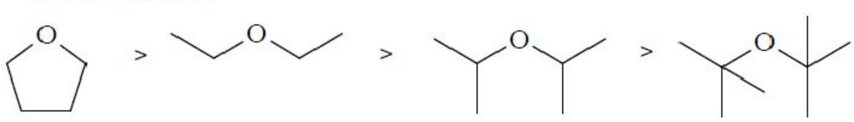
97. sec-Butyl alcohol and t-butyl alcohol can be distinguished by

A) Ceric ammonium nitrate B) I_2 / Na_2OH

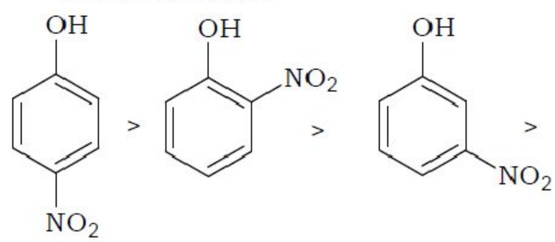
C) $K_2Cr_2O_7 / Con.H_2SO_4$ D) $Con.HCl / ahnyd.ZnCl_2$

98. Correct order(s) among the following is/are

Lewis Basicity

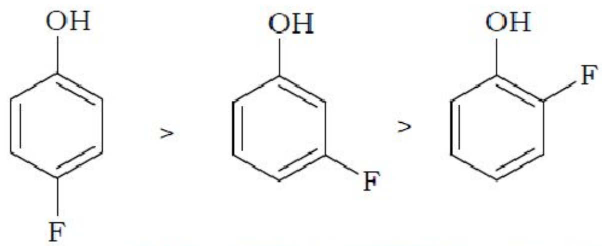
A) 

Acidic character



B)

Boiling Point



C)

D) **Reactivity towards cleavage of ethers** $HI > HBr > HCl$

99. Which of the following is correct statement

A) Resin and Rubber are natural polymers

B) rayon and cellulose nitrate are semi synthetic polymers

C) Synthetic fiber is nylon-6, 6 and synthetic rubber is buna-S

D) Bakelite and melamine are Bronched chain polymers

100. Correct matches is /are

A) Nylon-6,6, Nylon-6 $\longrightarrow$ Polyamides

B) Dacron or terylene $\longrightarrow$ Polyesters

C) 3 - hydroxy butanoic acid +3 - hydroxy pentanoic acid $\longrightarrow$ PHBV

D) Nylon-2-nylon-6 $\longrightarrow$ Alternating ester copolymer

101. Which of the following statements are correct?

A) Insulin is a peptide hormone

B) Progestrone is a female sex hormone

C) Vitamin-E is fat soluble vitamin

D) $\alpha - L$ Amino acids are monomer of proteins.

102. Which to the following antibiotics is/are the modification of penicillins ?

A) Ofloxacin B) Ampicillin C) Ranitidine D) Tetracycline

103. Which of the following antibiotics is effective against tuberculosis?

A) Chloromycetin B) Tetracycline C) Penicillin D) streptomycin

104. Which of the following pairs are bacteriostatic antibiotics?

A) Ofloxacin, tetracycline
B) Penicillin, Erythromycin
C) Erythromycin, chloramphenicol
D) Tetracycline, chloramphenicol

105. Which of the folliwng are the derivatives of barbituric acid?

A) Amytal
B) Luminal
C) Seconal
D) Mapromate

106. Select the correct statements from the following statements

A) On prolonged heating of glucose with HI forms n-Hexane

B) Glucose and Gluconic acid, both on oxidation with nitric and produce saccharic acid

C) Maltose is composed of α-glucose and β-Fructose

D) Sucrose is dextrorotatory but after hydrolysis gives dextrorotatory glucose $(+52.5°)$ and leavorotatory fructose $(-92.4°)$.

107. Which of the following statement are true

A) Except glycine, all other naturally occurring α-amino acids are optically active

B) Most naturally occurring amino acids have D-configuration

C) Insulin and albumins are the common examples of globular proteins

D) The proteins present in hair, wool, silk and in muscles are fibrous proteins

108. Which of the following is/are correct regarding DNA finger printing

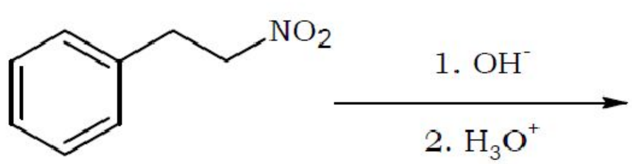
A) used in forensic laboratories for the identification of criminals

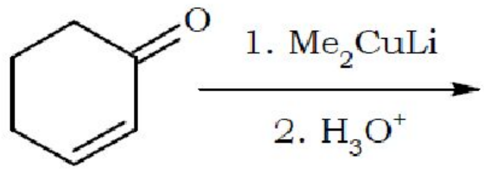
B) Used to determine paternity of an individual

C) Used to identify racial groups to rewrite biological evolution.

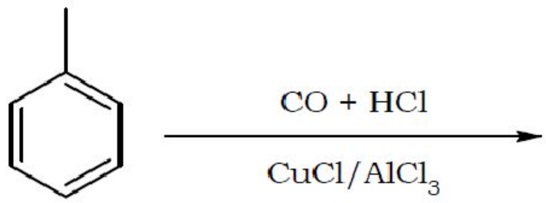
D) Used to identify the dead bodies in any accident by comparing the DNA's of parent (or) children

109. Which of the following reaction(s) give(s) an aldehyde or ketone as the principal organic product?

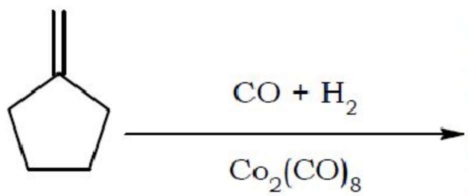
A) 



B)



C)



. D)

110. The boiling points of propan-1-ol and propanal are respectively 97 °C and 49 °C. Among the following condition(s) the ones which are effective for the conversion of propan-1-ol to propanal is/are

A) Distilling the alcohol with alkaline $KMnO_4$ at about 50 °C.

B) Distilling the alcohol with acidified dichromate at about 49 °C.

C) Distilling the alcohol with acidified dichromate at about 100 °C.

D) Distilling the alcohol with acidified dichromate at about 75 °C.

111. Organo copper and cadmium reagents react with less reactive acyl chlorides even though they don't react with the more reactive ketones because

A) R_2Cd and R_2CuLi are appreciably covalent and are not nucleophilic enough to add to ketones.

B) Cadmium and copper in these compounds complex with RCOCl, polarizing C – Cl bond of acylchloride making it more reactive acylium-like ion.

cC) Oxygen of carbonyl group of acyl halides complex with Cd and Cu making it more reactive.

D) Alpha hydrogen atoms of ketones are acidic rendering the reagents inactive by protonating them.

112. Which of the following sets of reagents can convert RCHO to RCDO

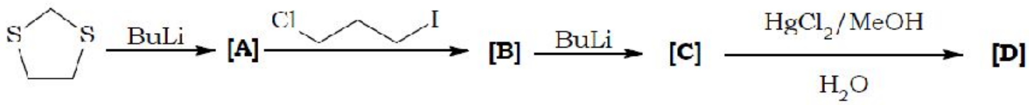
A) (i)$HS(CH_2)_3SH$, BF_3; (ii) BuLi; (iii) DCl; (iv)$HgCl_2/MeOH/H_2O$

B) (i)$HS(CH_2)_3SH$, BF_3; (ii) BuLi; (iii) D_2O; (iv)H_2/Ni

C) (i) $D_2/Pd\text{-}C$; (ii) PCC

D) (i)$HO(CH_2)_2OH/H^+$; (ii) NaOH; (iii) DCl; (iv)$HgCl_2/MeOH/H_2O$

113. Correct statement(s) regarding the following reaction sequence is/are



A) Formation of [B] as well as [C] involve nucleophilic substitution reactions.

B) [D] is a cyclic ketone.

C) [B] contains iodine atom.

D) [C] has no halogen atoms.

114. Which of the following reaction(s) give(s) an aldehyde as the product?

A) Treatment of RCN with 1 equivalent of $LiAlH_4$ at -50°C followed by hydrolysis.

(B) Treatment of RCN with MeMgCl followed by hydrolysis.

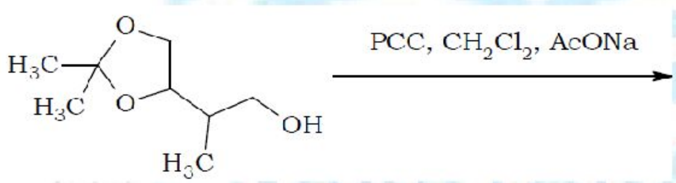
C) Treatment of HCN with MeMgBr followed by hydrolysis.

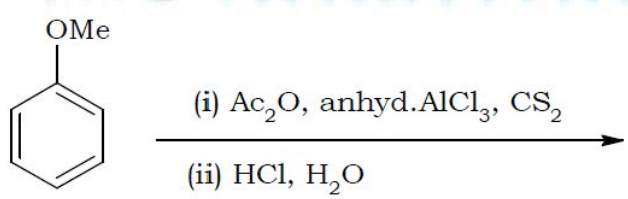
D) Treatment of resorcinol with HCN-HCl-$AlCl_3$ followed by hydrolysis.

115. Which of the following reaction give an aldehyde or ketone as the major product or exclusive product?

A) $$H_3C-\underset{\underset{OH}{|}}{CH}-C\equiv C-CH_2-CH_2-CH_3 \xrightarrow[0\ ^0C]{CrO_3/H_2SO_4/\text{acetone}}$$

B) 

C) 

D) 

116. Correct statement(s) regarding acetal formation reaction is/are

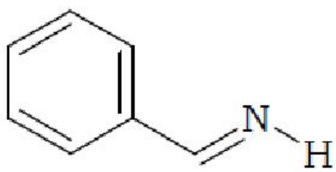
A) Hemiacetal formation is catalysed by acid but may also be catalysed by minute quantities of bases.

B) Ketones generally don't combine with simple alcohols but combine with glycols to give cyclic acetals.

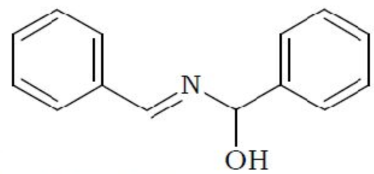
C) Acetals are stable in basic as well as dry acidic conditions but are cleaved readily by aqueous acids.

D) Hemiacetals are stable in dry acidic conditions and are cleaved by aqueous acids as well as bases.

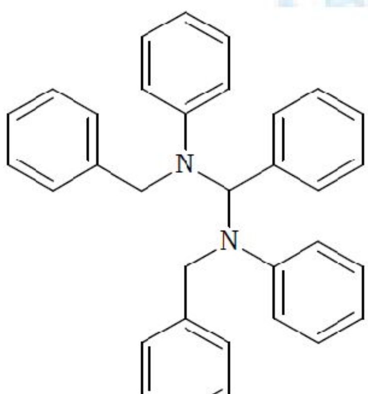
117. A stable product is isolated in the reaction between benzaldehyde and ammonia. The product is probably



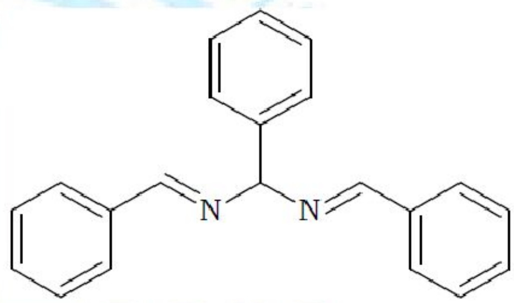
A)



B)



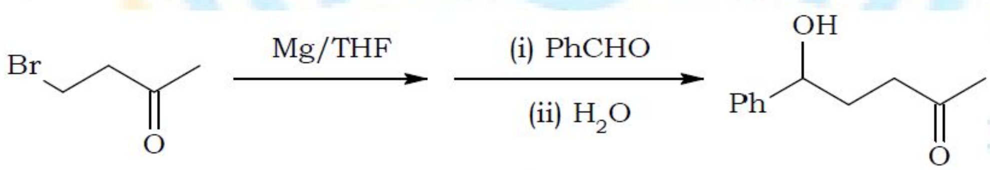
C)



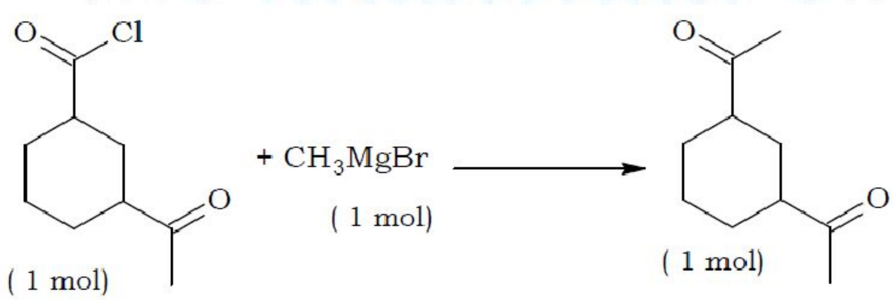
D)

118. Which of the following yield(s) the given product as the major/exclusive product?

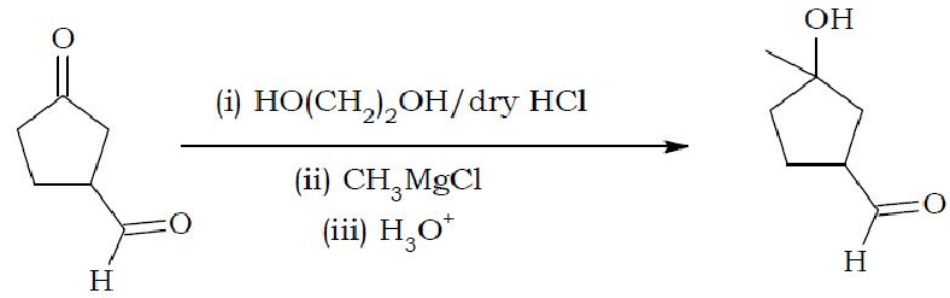
A)



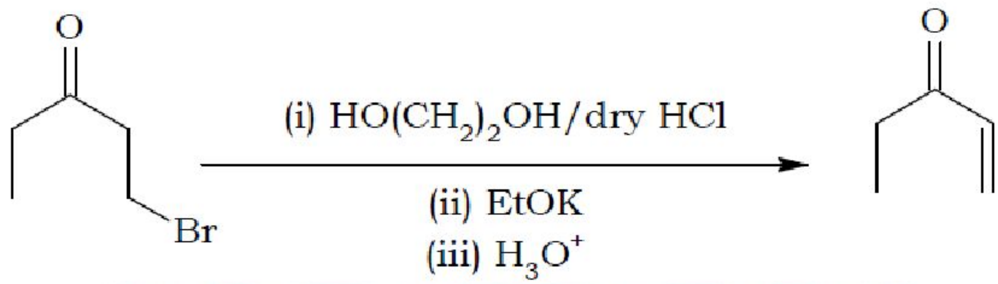
B)



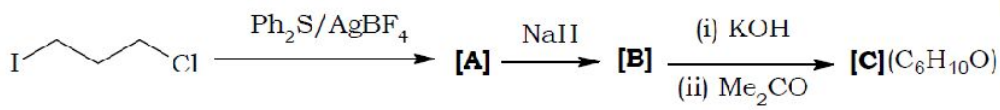
C)



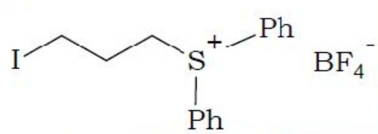
D)



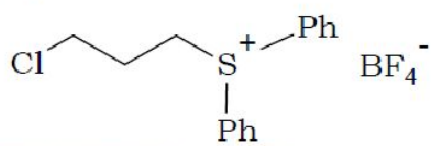
119. Missing compound(s) in the following reaction sequence is / are



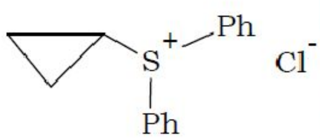
A) **[A]** is



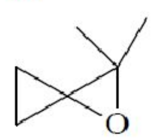
B) **[A]** is



C) **[B]** is



D) **[C]** is



120. During the addition of nucleophiles to carbonyl carbon, the nucleophile attacks the carbon in a way that the angle O – C – Nucleophile is about 107° in the transition state. This is to

A) provide maximum overlap between the filled orbital of nucleophile and empty antibonding orbital of carbonyl carbon.

B) minimise the repulsion between the pi cloud of carbonyl group and nucleophile as well as the alkyl groups bonded to carbonyl carbon.

(C) minimise the repulsion between the pi cloud of carbonyl group and the nucleophile as well as to provide appreciable overlap between the filled orbital of nucleophile and empty antibonding orbital of carbonyl carbon.

D) align HOMO of nucleophile and LUMO of carbonyl compounds parallel to each other.

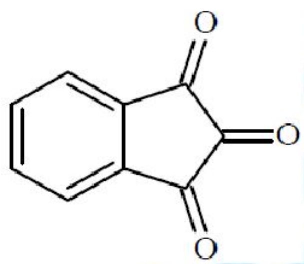
121. Correct statement(s) among the following regarding hydration of aldehydes and ketones is/are

A) $K_{eq}(HCHO) > K_{eq}(CH_3CHO) > K_{eq}(CH_3COCH_3)$

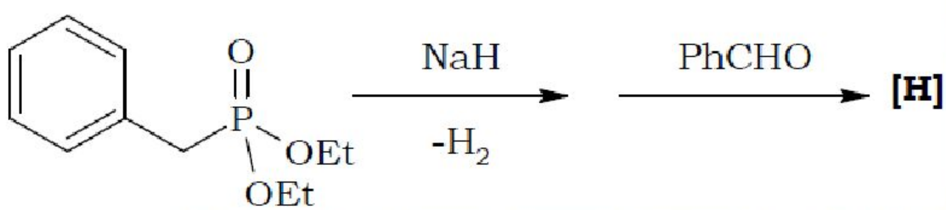
B) Cl_3CCHO forms a stable hydrate, the stability of which is attributed decrease in electrostatic repulsion due to hydration and intramolecular hydrogen bonding.

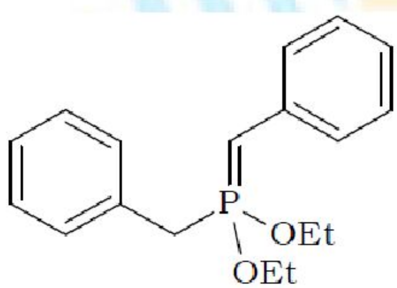
C) Stability of hydrates: cyclopropanaone > cyclobutanone > cyclopentanone

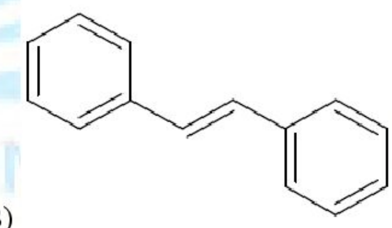
D) The following compound exists as a monohydrate, the middle carbonyl being the one receiving water.

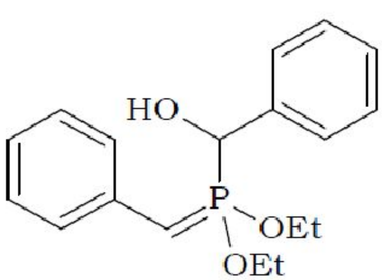


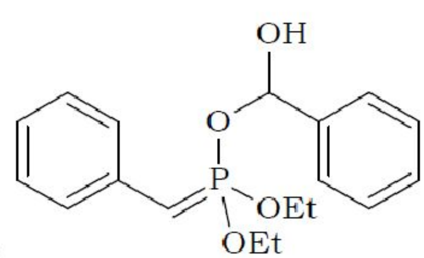
122. Compound [H] in the following reaction is



A) 

B) 

C) 

D) 

123. Which of the following factors decide whether reaction will be elimination versus substitution.

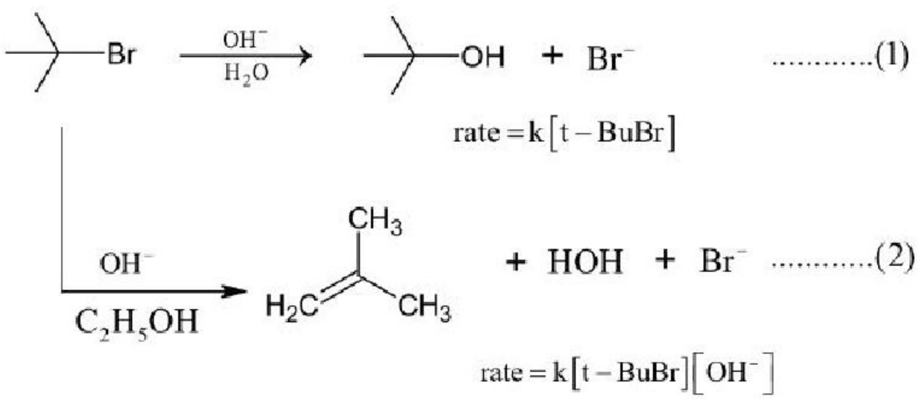
A) A Bulkier nucleophile will prefer to act as a base and abstracts a proton rather than approach a tetravalent carbon atom

B) High temperature generally are favourable for elimination reaction

C) A tertiary halide goes for S_N1 or elimination depending on stability of carbocation or the more substituted alkene formed as product

D) A secondary halide decides S_N2 or E_2 based on relative strength as a base/Nucleophile of the reagent

124. Consider the reaction sequence given below:



Which of the following statement is true?

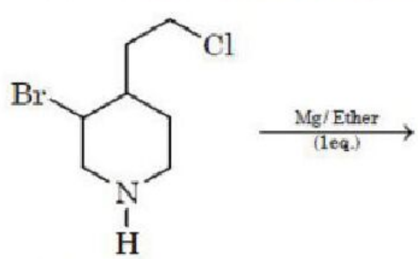
A) Changing the base for OH^- to ^-OR will have no effect on reaction (2)

B) Changing the concentration of base will have no effect on reaction (1)

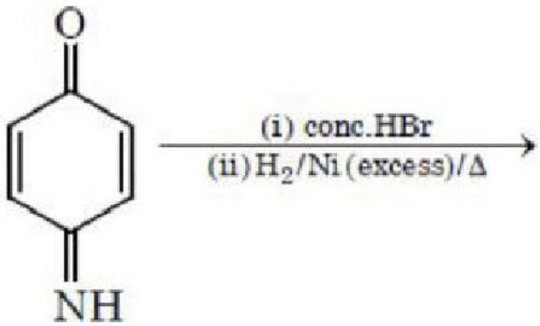
C) Doubling the concentration of base will double the rate of both the reactions

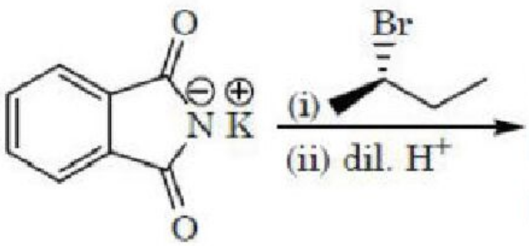
D) Doubling the concentration of base will double the rate of reactions (2)

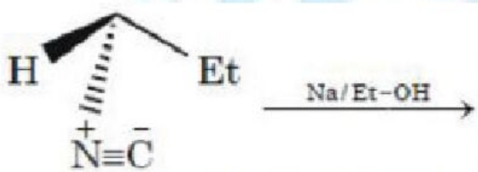
125. Identify which reaction(s) produce(s) single optically active Amine molecule as product.



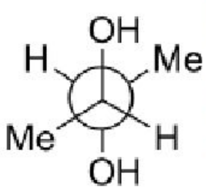
A) Optically pure

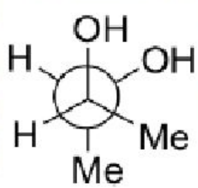
B) 

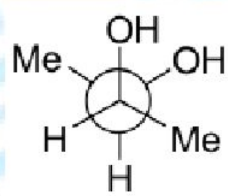
C) 

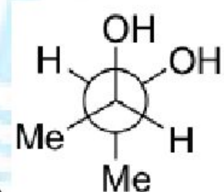
D) 

126. Among the following, the conformation that corresponds to the most stable conformation of meso-butane-2,3-diol is

A) 

B) 

C) 

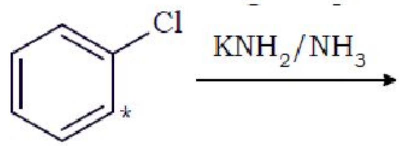
D) 

KEY

1	2	3	4	5	6	7	8	9	10
A	D	A	B	C	D	D	B	D	C
11	12	13	14	15	16	17	18	19	20
C	A	D	B	B	A	D	C	A	B
21	22	23	24	25	26	27	28	29	30
A	A	D	C	B	A	A	A	D	C
31	32	33	34	35	36	37	38	39	40
A	B	B	B	C	A	D	A	B	D
41	42	43	44	45	46	47	48	49	50
C	D	D	B	A	D	D	B	A	C
51	52	53	54	55	56	57	58	59	60
B	B	A	A	B	D	C	ABC	ACD	ABCD
61	62	63	64	65	66	67	68	69	70
BCD	ABCD	ABCD	ABD	ABC	ABC	ABCD	BC	AC	CD
71	72	73	74	75	76	77	78	79	80
AB	AD	ACD	BCD	ABC	C	AB	ABD	AB	BC
81	82	83	84	85	86	87	88	89	90
BCD	ABC	B,D	A	C	C	AD	ABCD	AB	C
91	92	93	94	95	96	97	98	99	100
ABCD	BC	D	ABD	ABD	AD	ABCD	ABCD	ABC	ABC
101	102	103	104	105	106	107	108	109	110
ABCD	B	D	CD	BC	ABD	ACD	ABCD	ABCD	D
111	112	113	114	115	116	117	118	119	120
AB	AC	ABD	AD	ABCD	ABCD	D	CD	BCD	C
121	122	123	124	125	126				
ABCD	B	ABCD	BD	ABCD	CD				

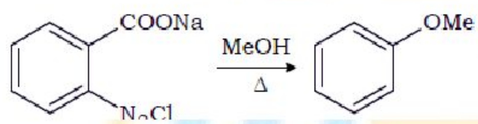
NUMERICAL TYPE QUESTIONS

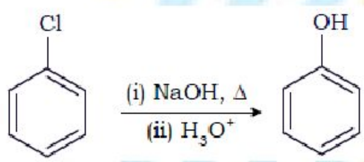
1. The number of organic products formed in the following reaction is

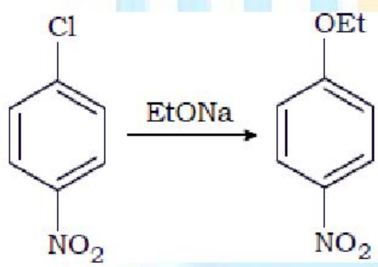


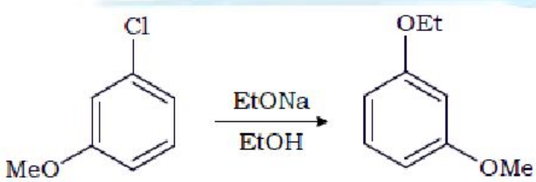
2. An alkane X has molecular formula C11H24. X on free radical chlorination gives three monochloro alkanes derivatives P, Q and R as positional isomers. Only Q is resolvable into enantiomers. P does not undergo E2 elimination reaction with ethanolic KOH while Q and R both give the same alkene with ethanolic KOH which is capable of showing geometrical isomerism. How many carbon atoms are present in the parent chain of X?

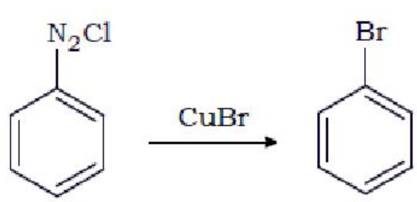
3. How many of the following reactions probably proceed via benzyne intermediate?

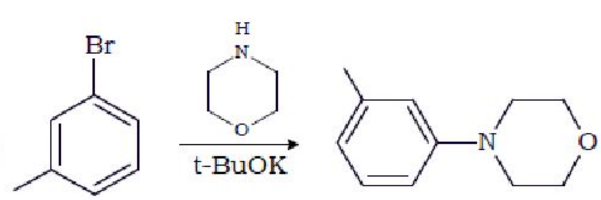
A) 

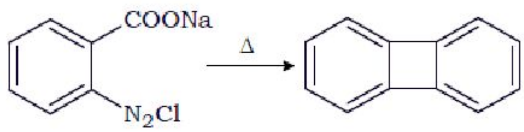
B) 

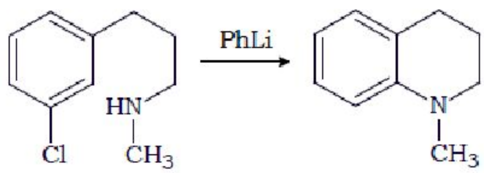
C) 

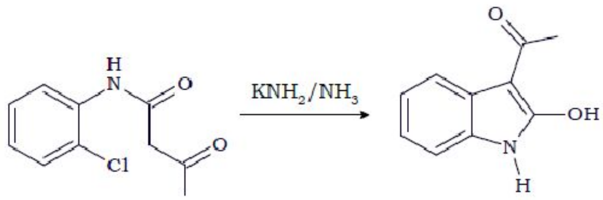
D) 

E) 

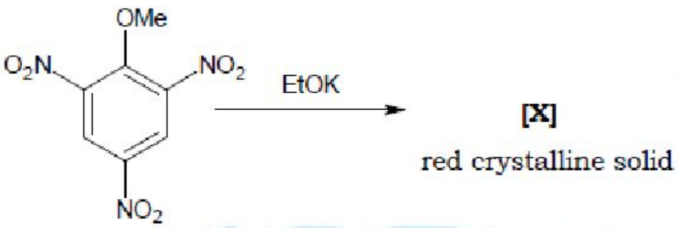
F) 

G) 

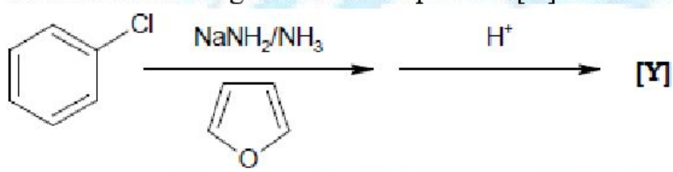
H) 

I) 

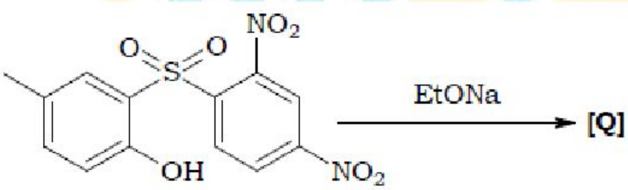
4. The molecular weight of the final product [X] of the following reaction is



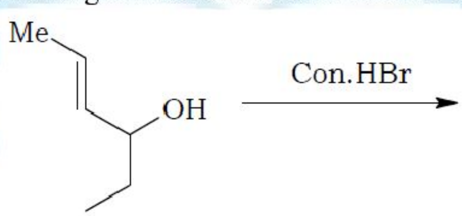
5. The molecular weight of the final product [Y] of the following reaction sequence is



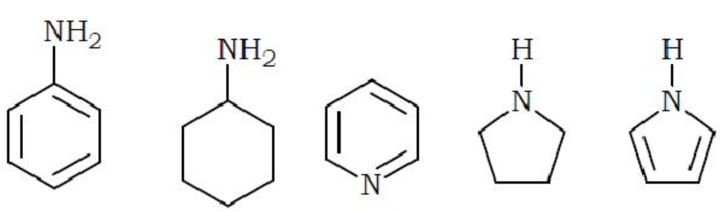
6. The number of oxygen atoms in the product [Q] of the following reaction is



7. The number of product(s) including stereoisomers formed in the following reaction is

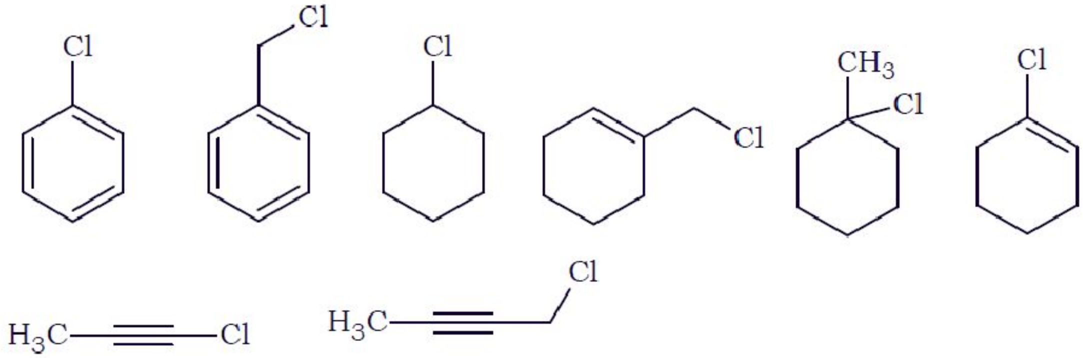


8. How many of the following nucleophiles are stronger than NH_3?

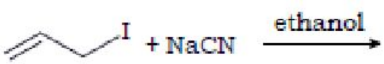


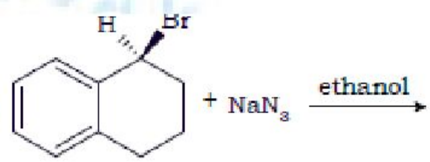
CH_3CONH_2 $NaNH_2$ NH_2NH_2 CH_3NH_2 CH_3CN

9. The number of compounds which give white precipitate with alcoholic $AgNO_2$ is

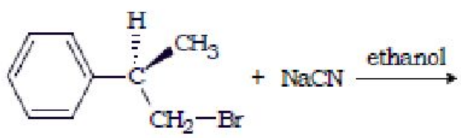


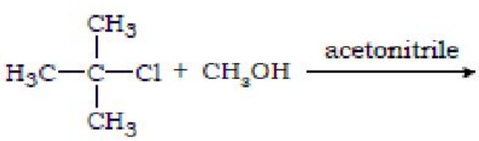
10. Among the following cases, the number of cases where there may not be any reaction i

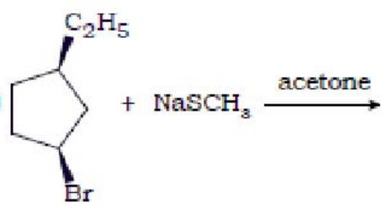
(i) 

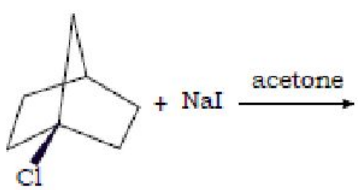
(ii) 

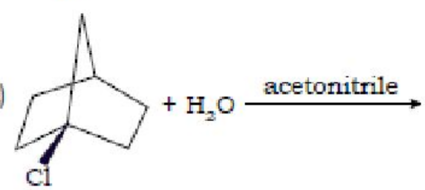
(iii) 

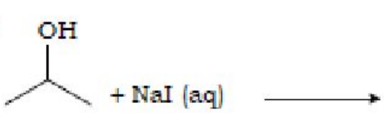
(iv) 

(v) 

(vi) 

(vii) 

(viii) 

(ix) 

11. Consider all isomeric alkenes, including stereoisomers of molecular weight 98. All these isomers are reacted independently with H2/Ni. Stereoisomers are also reacted separately. The total number of isomeric alkenes that give a racemic alkane is

12. How many of the following sequence of reagents would effect the given conversion?

a) $(i) HCl / ZnCl_2; (ii) Mg / Et_2O; (iii) HCHO; (iv) H_3O^+; (v) K_2Cr_2O_7 / H_2SO_4$

b) $(i) NaI / Con.H_2SO_4; (ii) NaCN; (iii) H_2O / H_2SO_4$

c) $(i) TsCl / Pyridine; (ii) NaI / acetone; (iii) Mg / Et_2O; (vi) CO_2; (v) H_3O^+$

d) $(i) Con.HBr; (ii) Mg / Et_2O; (iii) CH_3CHO; (vi) H_3O^+; (v) hot\ alk.KMnO_4;$

e) $(i) Con.H_2SO_4, \Delta; (ii) HBr; (iii) Li / Et_2O; (vi) CO_2; (v) H_3O^+$

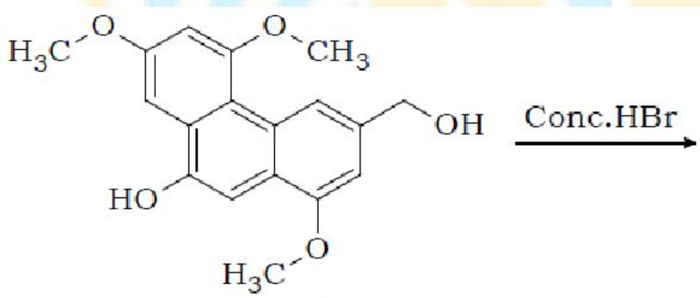
f) $(i) Con.H_3PO_4, \Delta; (ii) CO - HCl - AlCl_3\ (iii) H_3O^+$

g) $(i) Con.HBr; (ii) Mg / Et_2O\ (iii) CH_3CHO; (vi) H_3O^+; (v) Br_2 / NaOH; (vi) H_3O^+$

13. The number of compounds of molecular formula $C_8H_{10}O$ which give characteristic colour with $FeCl_3 / CHCl_3$ are

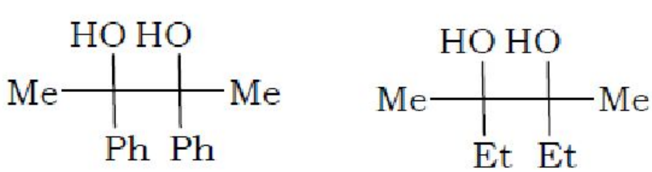
14. PCl_3 and an alcohol can react in x : y molar ratio to form corresponding alkyl halide and an oxyacid of basicity z. What is $\left(\frac{y}{x} + z\right)$?

15. How many moles of HBr is consumed in the following reaction?



One mole

16. A mixture of the two compounds below is treated with $Con.H_2SO_4$. Assuming that any group can migrate to the same extent as another group, the total number of pinacolones that are possible is



17. An excess of the racemic $CH_3CHClCOOH$ is allowed to react with (S)-2-methyl-1-butanol to form the ester, $CH_3CHClCOOCH_2CH(CH_3)CH_2CH_3$ and the reaction mixture is carefully distilled. The number of organic fractions that would be obtained is

18. The following reaction is carried out, and the products are separated by fractional distillation.

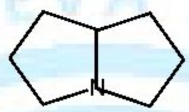
racemic $C_6H_5COCH\left(OH\right)C_6H_5 + H_2 / Ni \longrightarrow C_6H_5CHOHCH\left(OH\right)C_6H_5$

The number of fractions obtained is

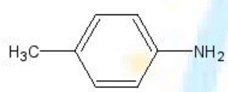
19. The number of alcohols formed by the solvolysis of 1-(chloromethyl)-4-methyl-1,3-cyclopentadiene is solvolyzed in aqueous acetone is

20. An unknown compound with molecular formula $C_9H_{12}O$ (i) does not decolourize Br_2 / CCl_4 (ii) is oxidized by hot $KMnO_4$ to give $PhCOOH$, and (iii) reacts with Na to give a colourless odourless gas. The number of compound(s) including stereoisomers which fit this scheme is

21. The number of Hoffmann's exhaustive methylation sequences required to get trimethyl amine from

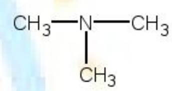


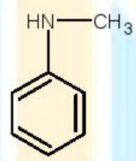
22. The number of compounds which give foul smell of isocyanide when they are heated with $CHCl_3$ in presence of alc. KOH.

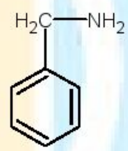


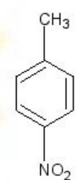
$CH_3 - CH_2 - NH_2$

$CH_3 - NH - CH_3$





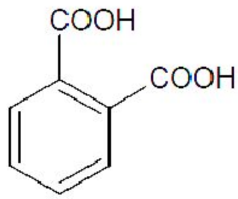
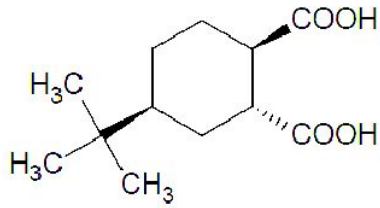
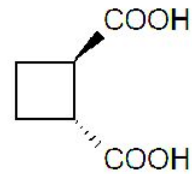


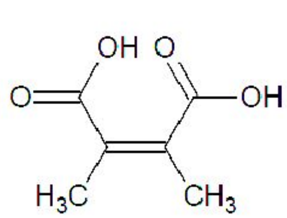
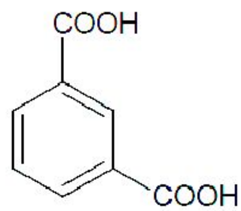
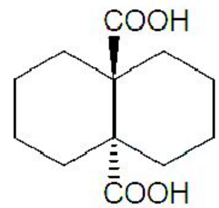


23. Number of possible 1^0 amine with molecule formula $C_5H_{13}N$. (excluding stereoisomers)

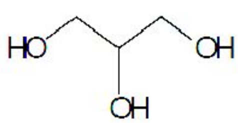
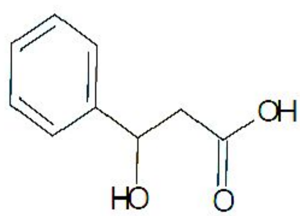
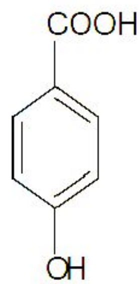
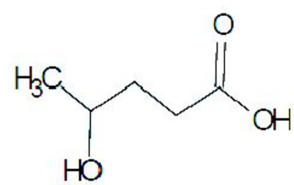
24. The number of moles of cyclohexylmethanol that can be oxidized to cyclohexanecarboxylate by 4 mol of KMnO4 in alkaline medium is.

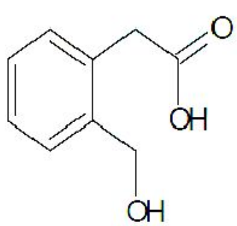
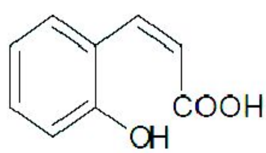
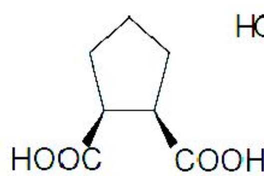
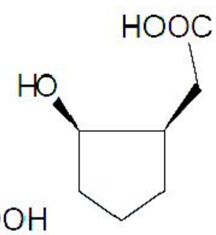
25. How many of the following dicarboxylic acids form cyclic anhydride just on heating?

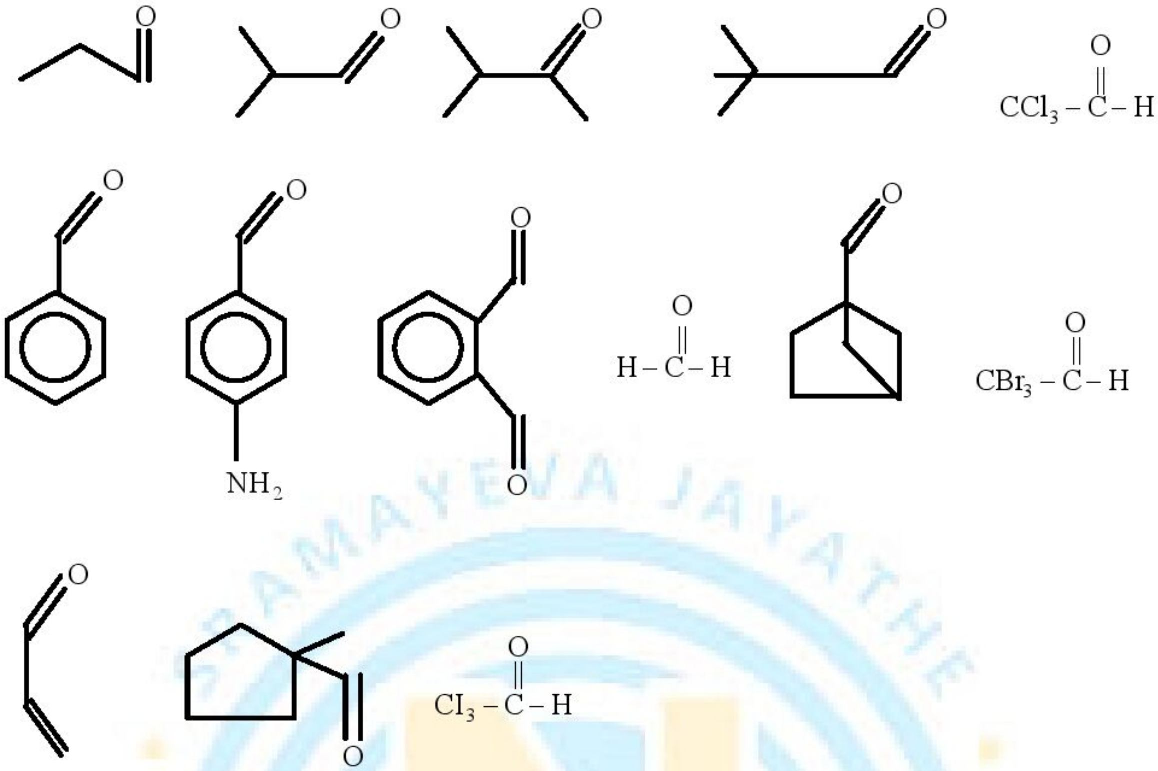




26. How many of the following compounds forms lactone on reaction with conc.H_2SO_4?

27. How many of the following compounds undergo self cannizero reaction (or) Internal cannizaro reaction.



28. How many chiral alcohols of the formula $C_5H_{12}O$ which can gives turbidity when treated anhydrous $ZnCl_2$ in conc. HCl at room temperature by SN_1 mechanism?

29. PCl_3 and an alcohol can react in **x : y** molar ratio to form corresponding alkyl halide and an oxyacid of basicity **z**. What is $\left(\frac{x}{x}+z\right)$

30. When phenol gets oxidized to 1,4-benzoquinone by acidified potassium dichromate, the number of moles of electrons transferred per mol of phenol is

31. How many of the following statements is/are true

a) Buna-N is a copolymer of 1,3-butadiene and acrylonitrile

b) Buna-S is a copolymer of 1,3-butadiene and styrene

c) Terylene and Bakelite are condensation polymers

d) Polythene and polyvinyl chloride are linear polymers

e) PHBV is a co-polymer of Hexamethylenediamine

f) Low density polythene is a branched chain polymer

g) Natural rubber becomes soft at low temperature (<283K) and brittle at high temperature (>335K)

h) In the manufacture of tyre rubber 5% of ‘S’ is used as a cross linking agent

i) Novolac on heating with formaldehyde undergoes cross linking to form an infusible solid mass called glyptal

32. How many of the following statements are correct

a) Low density polythene is chemically inert and tough but flexible and a poor conductor of electricity

b) High density polythene is used for manufacturing buckets, bottles and Dust bins.

c) The catalyst used in the polymerization of acrylonitrile is a peroxide

d) Terylene is formed by the polymerization of Glyoxal and terephthalic acid

e) Nylon-6,6 is obtained by heating caprolactum with water at a high temperature at 533-543 K

f) Dacron fibre is crease resistant and is used in blending with cotton, Wool fibres and in safety helmets

g) PHBV is used in specialty packaging orthopaedic devices and in controlled release of drugs.

h) Ali phatic polyesters not biodegradable polymers

33. How many of the following are incorrect statements

a) Nylon-2-nylon-6 is an alternating polyamide copolymer of glycine and amino caproic acid

b) Nylon-2- nylon-6 is non biodegradable polymer

c) PHBV undergoes bacterial degradation in the environment

d) Neoprene is formed by the free radical polymerization of 2- Methyl-1, 3- butadiene

e) High density polythene is obtained by the polymerisation of ethene in the presence of wilkinson's catalyst

f) Polychloroprene has superior resistance to vegetable and mineral oils.

34. How many of the following are antibiotics?

Chloramphenicol, chloroxylenol, streptomycin, saccharin, primaquin, penicillin, ampicillin, equanil, tetracyclin.

35. How many of the following are antihistamine drugs?

Salvarsen, streptomycin, promethazine, iproniazid, terfenadine, diclofenac sodium, bromopheniramine, lansoprazole, diphenyl hydramine.

36. How many of the following are tranquillizers?

Valium, seconal, diphenhydramine, primaquin, luminal, veronal, ranitidine, barbituric acid, mifepristone, equanil.

37. The number of correct statements

A) Antibiotics which kill or inhibit a wide range of Gram – positive and Gram – negative bacteria are said to be broad spectrum antibiotics

B) Ortho – sulphobenzimide is also called saccharin

C) Chloroxylenol is an anti fertility drug

D) Sodium benzoate acts as food preservative

38. How many of the following are correct

A) Anionic detergents are sodium salts of sulphonated long chain alcohols or hydrocarbons

B) Cationic detergents are equaternary ammonium salts of amines with acetates, chlorides or bromides as anions

C) detergent produced by the action of stearic acid with polyethylene glycol is a cationic detergent

D) All

39. How many of the following are incorrect statements

A) Drugs that bind to the receptor site and inhibit its natural function are called antagonists.

B) The drugs that mimic the natural messenger by switching on the receptor are called agonists.

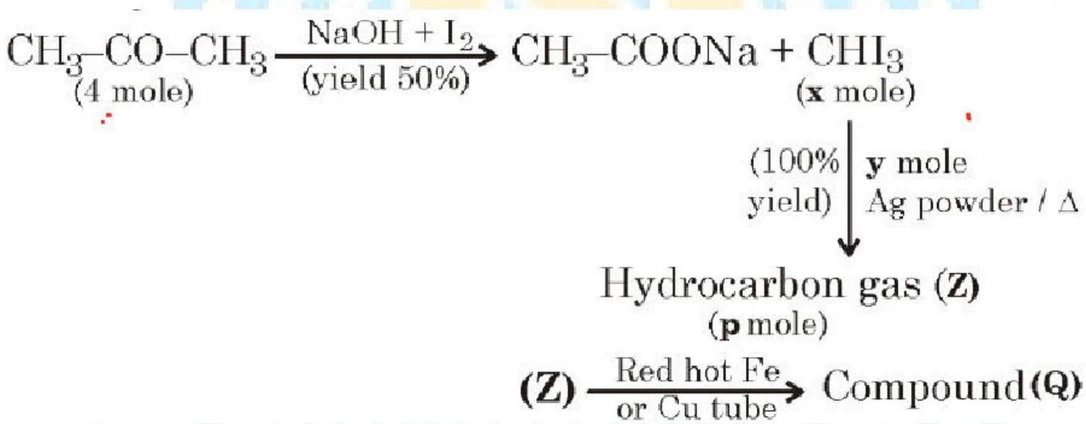
C) Receptors are proteins that are embedded in cell membranes

D) Drugs compete with the natural substrate for their attachment on the active sites of enzymes are called enzyme inhibitors.

STEM TYPE QUESTIONS

40 - 41

Consider given reactions:



are t = degree unsaturation of compound Q

40. Value of (x + t) is…..

41. Value of y + p is.

PARAGRAPH TYPE

PARAGRAPH – (42 – 44)

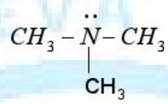
Compound A, ($C_5H_{11}NO$) is a neutral compound when this was heated in presence of NaOH, it forms a salt B and C. This C when reacts with acetyl chloride forms D. (C_4H_9NO). C also reacts with HNO_2 to give yellow oily solution. This C was also reacting with Hinsberg reagent but insoluble in alkali.

42. The compound C in the above sequence is

A) $CH_3-\ddot{N}H-CH_3$

B) $CH_3-CH_2-NH_2$

C) $CH_3-NH-CH_2-CH_3$

D) 

43. The compound ‘C’

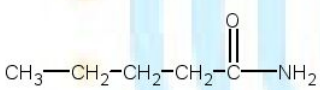
A) Is less basic than methyl amine & trimethyl amine in aq. Solution

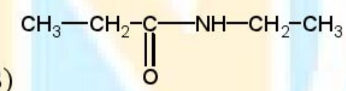
B) Gives carbyl amine reaction

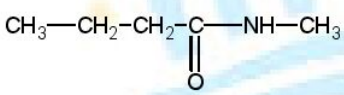
C) Is more basic than methyl amine & trimethyl amine is aq. Solution

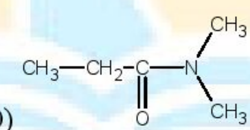
D) Can take 3 moles of methyl iodide to form quaternary ammonium salt.

44. Compound A is

A) 

B) 

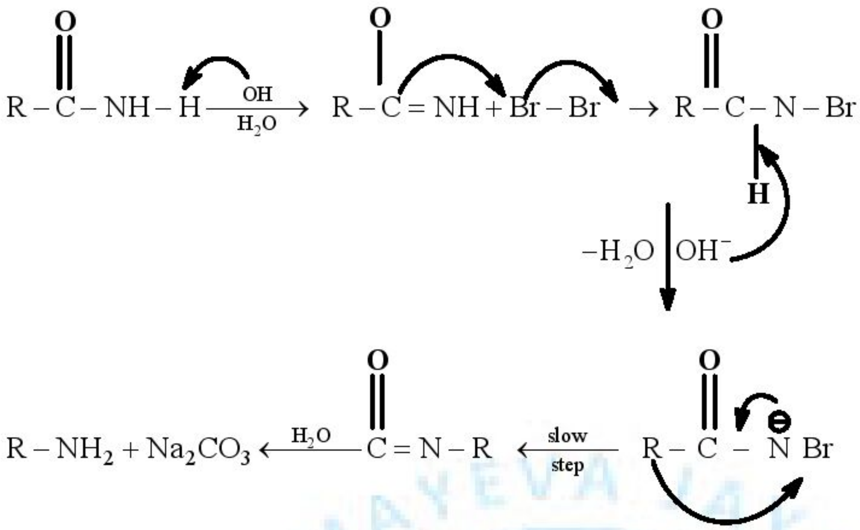
C) 

D) 

PASSAGE - 45 – 47

The conversion of an amide by action of NaOH and Br_2 to primary amine that has one carbon less than the starting amide known as Hofmann-Bromoamide reaction

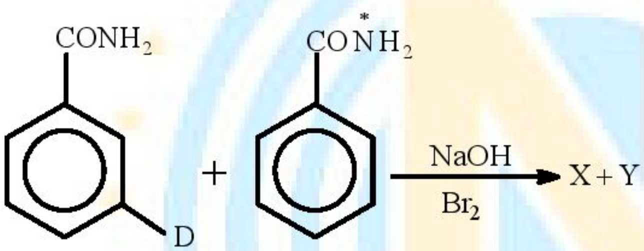
$$\underset{\text{Mechanism}}{H_3C-\overset{\overset{\displaystyle O}{\|}}{C}-NH_2}\xrightarrow[\text{or NaOBr}]{Br_2+NaOH}R-NH_2+NaBr+Na_2CO_3$$



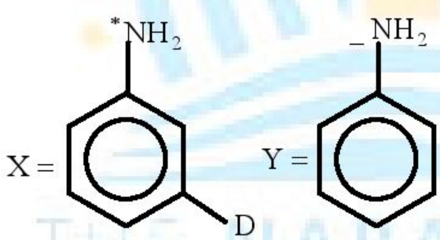
45. Number of moles of NaOH consumed in above reaction

A) 1 B) 2 C) 3 D) 4

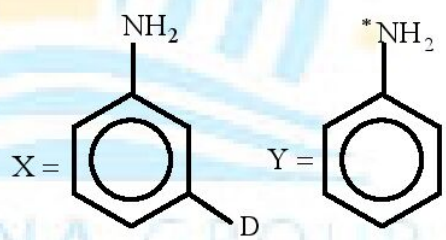
46.



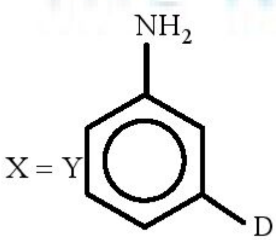
Find X and Y



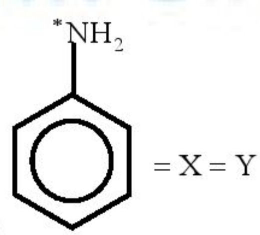
A)



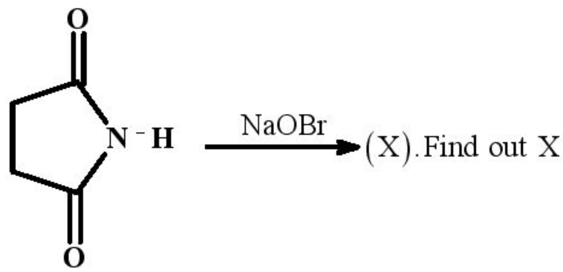
B)

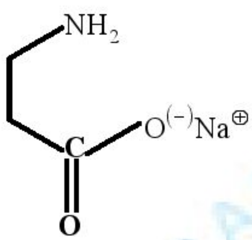
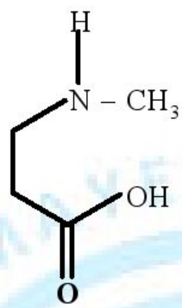
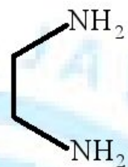


C)



D)

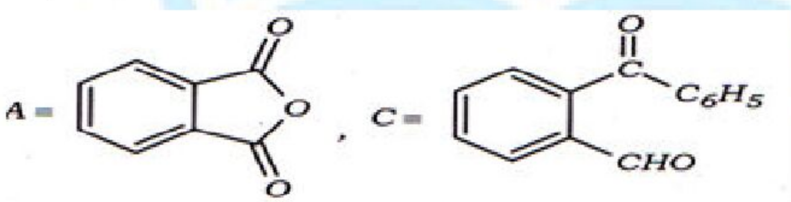
47. 

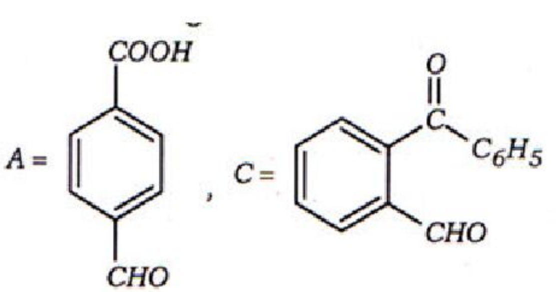
A)  B)  C)  D) All of these

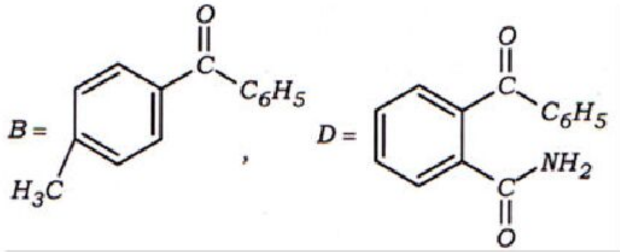
Passage – 48 – 49

An organic compound A, $C_8H_4O_3$ in dry benzene in the presence of anhydrous $AlCl_3$ gives compound B. The compound B on treatment with PCl_5. Followed by reaction with $H_2/Pd/BaSO_4$ gives compound C. which on react with hydrazine $(NH_2 - NH_2)$ gives cyclized compound D. $(C_{14}H_{10}N_2)$.

48. Identify the correct compounds from the following

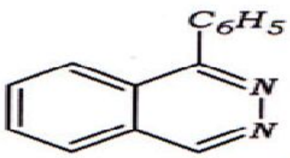
A) 

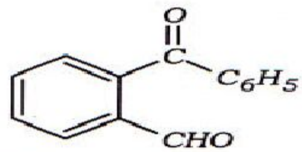
B) 

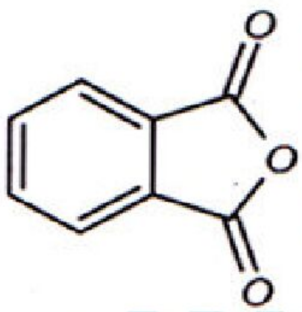
C) 

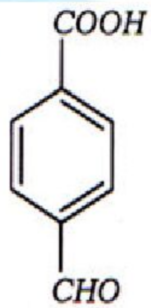
D) All the above

49. Compound D is

A) 

B) 

C) 

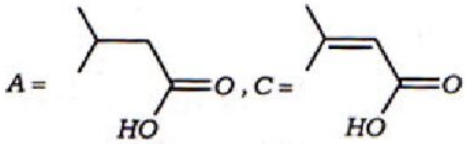
D) 

Passage – 50 – 51

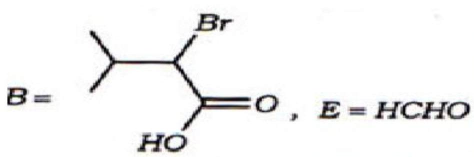
Organic acid (A), $C_5H_{10}O_2$ reacts with Br_2 in the presence of phorphorus to give compound (B). (B) contains an asymmetric carbon atom and yields (C). On dehydrobromination compound (C) does not show geometric isomerism and on decarboxylation, gives an alkene (D). which on oxidative ozonolysis gives (E) and (F). Compound (E) gives a positive Schiffs test but (F) does not give.

50. Which of the following combinations represent correct compounds

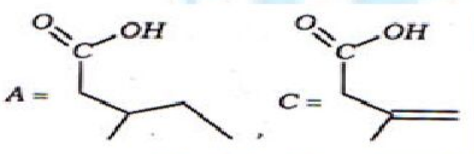
A)



B)

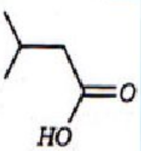
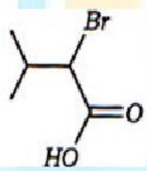
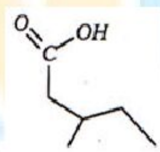
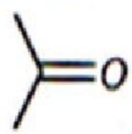


C)



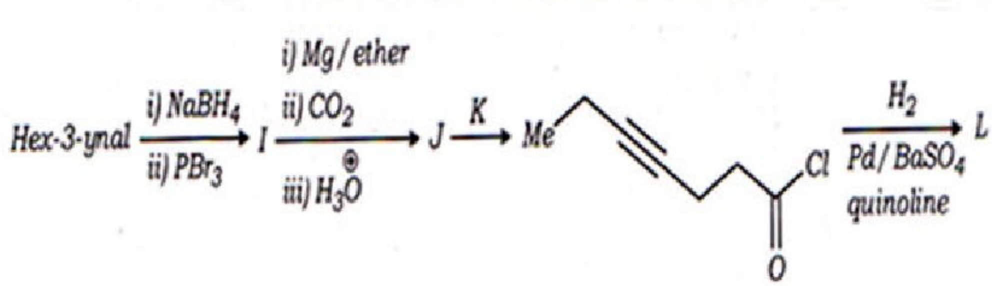
D) All the above

51. Compound (E) is

A)  B)  C)  D) 

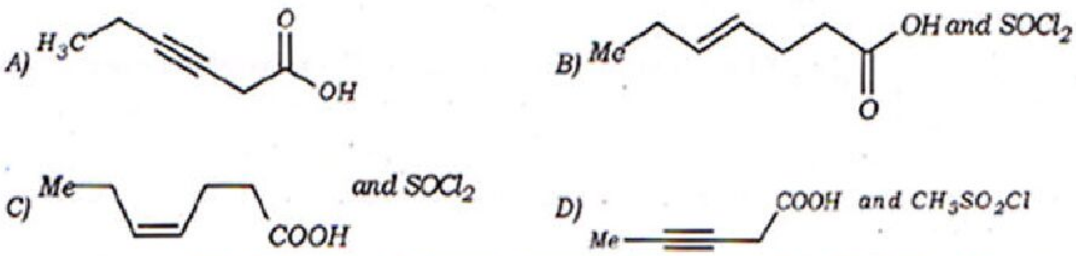
Passage – 52 - 54

In the following reaction sequence, products I, J and L are formed. K represents a reagent



52. The structure of the product 'I' is

53. The structures of compounds J and K, respectively, are



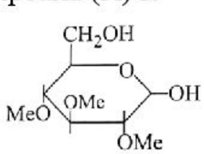
54. The structure of product L is

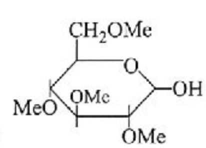
Passage – 55 – 57

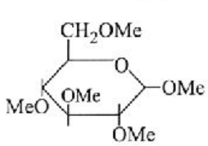
Consider the following sequence of reactions $D(+)$ glucopyranose $\xrightarrow{CH_3OH/H^{\oplus}} (A)$

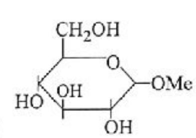
$\xrightarrow{HIO_4} (B)+(C) \xrightarrow[(ii)\,H_3O^{\oplus}]{(i)\,Br_2\ water} (D)+(E)$

55. Compound (A) is

A) 

B) 

C) 

D) 

56. Compound (B) gives yellow precipitate with Fehling solution. Compound (B) is

A) $HCOOH$ B) $HCHO$

C) Both (A) and (B) D) CH_3-CHO

57. (D) and € respectively are

A) $\begin{array}{c} COOH \\ | \\ COOH \end{array}$ and $\begin{array}{c} COOH \\ | \\ H-C-OH \\ | \\ COOH \end{array}$

B) $\begin{array}{c} CHO \\ | \\ COOH \end{array}$ and $\begin{array}{c} COOH \\ | \\ H-C-OH \\ | \\ COOH \end{array}$

C) $\begin{array}{c} CHO \\ | \\ CHO \end{array}$ and $\begin{array}{c} COOH \\ | \\ H-C-OH \\ | \\ CH_2OH \end{array}$

D) $\begin{array}{c} CHO \\ | \\ COOH \end{array}$ and $\begin{array}{c} COOH \\ | \\ H-C-OH \\ | \\ CH_2OH \end{array}$

Passage 58 - 60

Antibiotics are the chemical substances which are produced by micro organisms like bacteria, fungi and moulds. Antibiotics can inhibit the growth or even destroy other microorganisms. Now-a-days, synthetic antibiotics are also available. The first successful antibiotic produced was penicillin. The antibiotics may be either bacteriocidal (kills the organism in the body) or bacteriostatic (inhibits the growth of organism). Ampicillin and amoxycillin are modified antibiotics. Broad spectrum antibiotics are effective against several types of harmful microorganisms.

58. Chloramphenicol is

A) antipyretic B) broad spectrum antibiotic C) azo dye D) tranquillizer

59. Which of the following is/are not an antibiotic?

A) Chloramphenicol B) Sulphadiazine C) Penicillin D) All

60. Which of the following antibiotics is bacteriostatic?

A) Penicillin B) Ofloxacin

C) aminoglycosiders D) Erythromycin

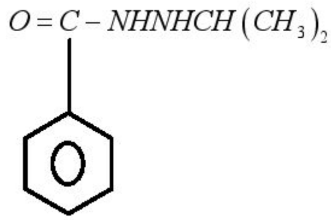
PASSAGE – 61 – 62

A major break through in the treatment of hyper acidity came through the discovery according to which a chemical histamine stimulates the secretion of pepsin and hydrochloric acid in the stomach. The drug cimetidine was designed to prevent the interaction of histamine with the receptors present in the stomach wall. The importance of the drug was so much that it retained the largest selling drug in the world until another drug ranitidine (Zantac) was discovered.

61. Which of the following is antacid

A) seldane B) Brompheniramine C) Cimetidine D) All

62. Which of the following structure represents Histamine

A) 

B) 

C) 

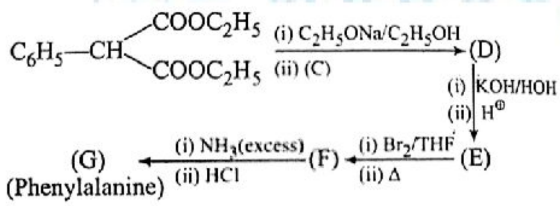
D) 

Passage – 63 – 64

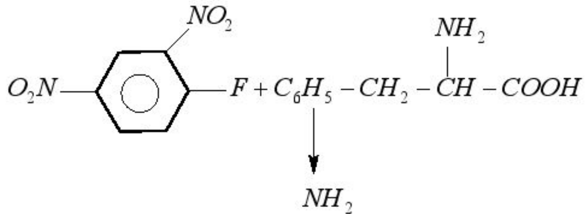
Compound (A) is soluble in $NaHCO_3$. On oxidation with $KMnO_4/H^{\oplus}$ it gives benzoic acid. Compound

(A) gives the following sequence of reactions:

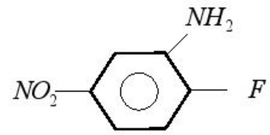
(A) $(A)\xrightarrow[\Delta]{Soda\,lime}(B)\xrightarrow[\Delta]{Soda\,lime}(C)$

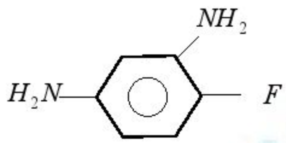


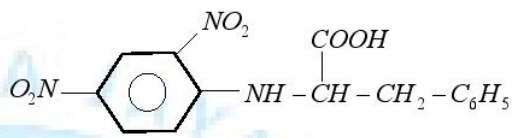
63. In the given reaction sequence product (P) is



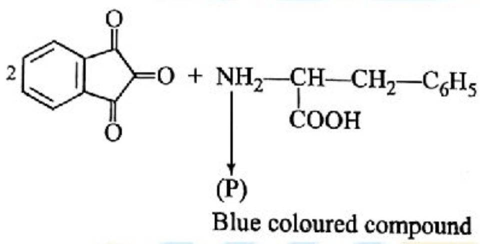
A) $C_6H_5-CH_2-\overset{\displaystyle NH_2}{\overset{|}{CH_2}}$

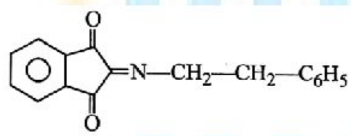
B) 

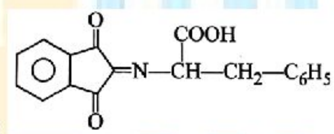
C) 

D) 

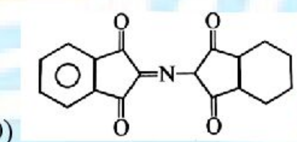
64. Product (P) in the given reaction sequence is



A) 

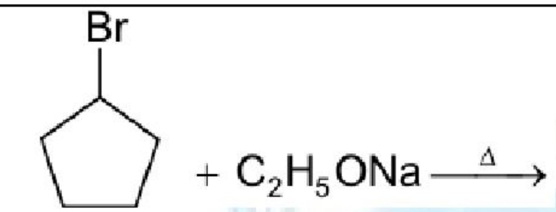
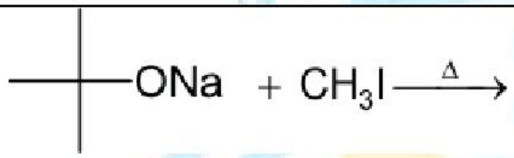
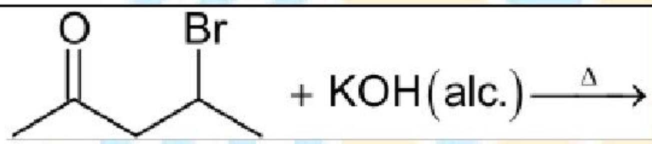
B) 

C) $C_6H_5-CH_2-CHO$

D) 

MATCHING TYPE

65. Match the reactions given List I with their corresponding characteristics in List II and select the correct option from the codes given below

List I		List II	
I)		P)	Major product is formed via E1cb mechanism
II)		Q)	Major product is formed via E_2 or E_2' mechanism
III)		R)	Reaction proceeds with the formation of transition state
IV)		S)	Carbanion is formed as intermediate
		T)	Major product is formed via S_N2 mechanism

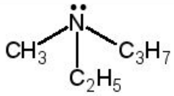
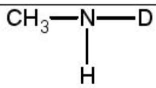
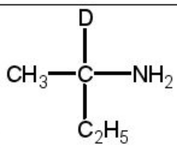
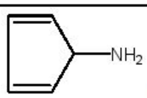
A) I – Q, R; II – R, T; III – P, S; IV – Q, S

B) I – Q, R; II – R, T; III – P, S; IV – P, S

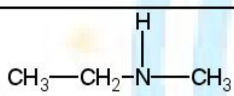
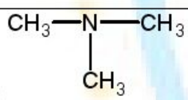
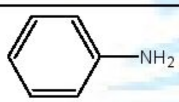
C) I – R, T; II – Q, T; III – P, T; IV – P, S

D) I – Q, R; II – R, T; III – P, S; IV – Q, R

66. Match the following

Column – I	Column – II
A) 	P) Optically inactive
B) 	Q) one chiral atom is present
C) 	R) Aliphatic 1^0 amine
D) 	S) aromatic primary amine

67. Match the following

List – I	List – II
A) $CH_3CH_2CH_2\ NH_2$	P) Treatment of $NaNO_2$, HCl gives N-Nitroso compound
B) 	Q) Treatment of $NaNO_2$, HCl gives diazonium chloride
C) 	R) Treatment of excess CH_3I followed by AgOH and heat gives out alkene
D) 	S) Does not show Hofmann mustard oil reaction

68. Match the following

Column-I		Column-II	
a	$\alpha-D(+)$ glucopyranose and $\beta-D(+)$ glucopyranose	P	Functional isomers
b	Glucose and Mannose	Q	Hemiacetal
c	Glucose and galactose	R	Epimers
d	Glucose and fructose	S	Anomers
		T	Mutarotation

69. Match the following

Column-I		Column-II	
a	Cellulose	P	Biopolymer
b	Starch	Q	glucose
c	Protein	R	$\alpha - L -$ Amino acids
d	Nucleic acid	S	Nucleotides

70. Match the following

Column-I		Column-II	
a	Amylose	P	$\alpha -$Helical structure
b	Amylopectin	Q	Double helix
c	DNA	R	Linear structure
d	Protein	S	Branch chain structure
		T	Biopolymer

71. MATCH THE FOLLOWING

LIST – I		LIST – II	
A)	Antacid	p)	Meprobamate
B)	Antihistamine	q)	Aspirin
C)	Non – Narcotic	r)	Ranitidine
D)	Tranquilizer	s)	Terfenadine

MATCHING THREE COLUMN TYPE

Column 1,2 and 3 contain many reagents and conditions.		
(I) HBr	(i) $NaBH_4$	(P) $(a) CH_3CHO$; $(b) H_3O^+$; $(c) I_2 / NaOH$; $(d) H_3O^+$
(II) NBS / CCl_4	(ii) Mg / Et_2O	(Q) $(a) HCHO$; $(b) H_2CrO_4 / acetone$
(III) $Hg(OA)_2, H_2O$	(iii) PDC	(R) $(a) NaOCl$ $(b) H_3O^+$
(VI) $(a) B_2H_6 / THF$ $(b) H_2O_2 / NaOH$	(iv) $(a) MeMgBr$ $(b) H_3O^+$	(S) $(a) CO_2$; $(b) H_3O^+$

72. Which of the following sequence(s) of reagents may convert cis-Pent-2-ene in to a mixture of three or more isomeric carboxylic acids?

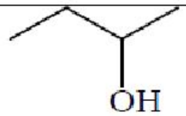
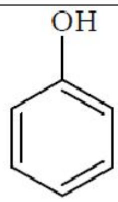
A) (I) (ii) (S) B) (II) (ii) (S) C) (III) (i) (S) D) (IV) (ii) (P)

73. Which of the following combination of reagents convert propene into but-3-enoic acid?

A) (II) (ii) (Q) B) (III) (iv) (R) C) (IV) (ii) (P) D) (II) (ii) (P)

74. Propyne gets converted in to propanoic acid by

A) (II) (i) (R) B) (IV) (iv) (R) C) (III) (ii) (S) D) (IV) (i) (R)

Column 1,2 and 3 contain reactants, reagents and the visual changes exhibited when they are mixed.		
(I) 	(i) Br_2 / CCl_4	(P) Colour change
(II) 	(ii) $K_2Cr_2O_7 / H_2SO_4$	(Q) Precipitate formation
(III) 	(iii) $\left[Ag\left(NH_3\right)_2\right]Cl$	(R) evolution of gas
(VI) 	(iv) Blue litmus solution	(S) characteristic odour

75. The CORRECT combination(s) among the following is/are

A) (I) (i) (P) B) (II) (ii) (Q) C) (III) (i) (R) D) (IV) (iii) (Q)

76. The INCORRECT combination(s) among the following is

A) (I) (ii) (P) B) (II) (i) (P) C) (III) (iv) (P) D) (IV) (i) (R)

77. The only CORRECT combination among the following is

A) (I) (iv) (P) B) (II) (i) (Q) C) (III) (ii) (S) D) (IV) (i) (S)

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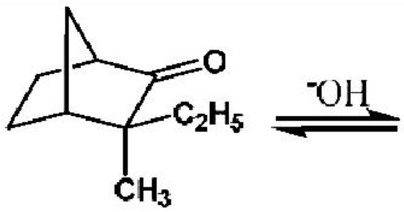
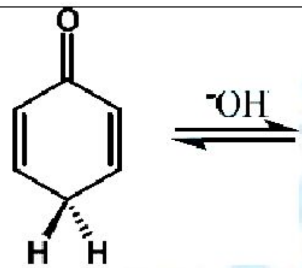
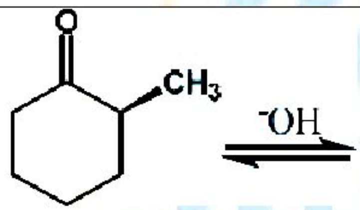
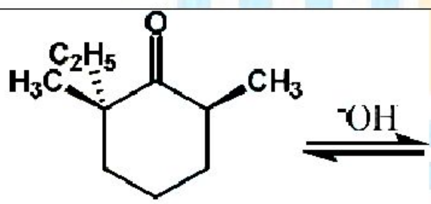
Answer 78, 79 and 80 by appropriately matching the information given in the three columns of the following table.

Column I		Column II		Column III	
I)		(i)	Shows keto-enol tautomerism	P)	Optically pure compound of column-I gives diastereomers on tautomerisation
II)		(ii)	Do not Shows ketoenol tautomerism	Q)	Optically pure compound of column-I gives enantiomers on tautomerisation
III)		(iii)	Example of 1,3-traid system	R)	Either keto form is predominent in liquid state or only keto form exist
IV)		(iv)	Example of 1,5-traid system	S)	Enol form is predominent in liquid state

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78. Correct combination for the above three column table is?

A) I-(i)-R B) I-(i)-S C) I-(ii)-R D) I-(ii)-S

79. Incorrect combination of the above three column table is?

A) III-(i)-P B) III-(iii)-S C) IV-(i)-Q D) All (A,B,C)

80. Correct combinaion of above three column table?

A) III-(i)-Q B) IV-(i)-P C) IV-(iii)-R D) All (A,B,C)

Answer 81, 82 and 83 by appropriately matching the information given in the three columns of the following table.

Column I		Column II		Column III	
I)	Propiophenone	(i)	$NaOH/Br_2/H^{\oplus}$	P)	Condensation
II)	Acetophenone	(ii)	Br_2/H^+	Q)	Formylation reaction
III)	Benzaldehyde	(iii)	$(CH_3CO)_2O/CH_3COOK/H^{\oplus}$	R)	Substitution
IV)	Toluene	(iv)	$CrO_{3/}(AcO)_2O$	S)	Haloform

81. The only correct combination in which the reaction proceeds enol intermediate is

A) I-(i)-P B) I-(i)-Q C) I-(ii)-R D) I-(ii)-S

82. For the synthesis of benzoic acid, the only correct combination is

A) III-(iv)-R B) III-(iii)-S C) II-(i)-S D) All (A,B,C)

83. The only correct combination that gives two different (structurally) carboxylic acids is

A) III-(i)-Q B) III-(iii)-P C) IV-(iv)-Q D) All (A,B,C)

MATCH THE FOLLOWING

LIST – I		LIST – II		LIST – III	
A)	Glyptal	1)	Vinyl benzene	p)	Manufacture of paints and lacquers.
B)	Bakelite	2)	Ethylene Glycol puthalic acid	q)	For making utensils and computer discs
C)	Polystyrene	3)	Phenol formaldehyde	r)	Manufacture of toys and t.v. cabinets

84. Which is correct combination

A) A – 2 – p B) B – 1 – q C) C – 2 – r D) A – 3 – r

85. Which is incorrect combination

A) A – 2 – p B) B – 3 – q C) C – 1 – r D) B – 2 – p

KEY

1	2	3	4	5	6	7	8	9	10
3	**7**	**7**	**327**	**308**	**7**	**8**	**5**	**5**	**3**
11	12	13	14	15	16	17	18	19	20
9	**5**	**9**	**5**	**4**	**4**	**3**	**2**	**5**	**8**
21	22	23	24	25	26	27	28	29	30
3	**4**	**8**	**4**	**2**	**4**	**8**	**6**	**5**	**4**
31	32	33	34	35	36	37	38	39	40
6	**5**	**3**	**5**	**4**	7	**3**	**2**	**3**	**6**
41	42	43	44	45	46	47	48	49	50
1	**A**	**C**	**D**	**D**	**B**	**A**	**A**	**A**	**A**
51	52	53	54	55	56	57	58	59	60
D	**D**	**A**	**C**	**D**	**A**	**D**	**B**	**B**	**D**
61	62	63	64	65	66	67	68	69	70
C	**D**	**D**	**D**	**D**	**A– P,Q B-P,Q,R C – Q,R D – P,R**	**A– Q,R B – P,R C – S D – Q**	**a-Q,S,T, b – R, T, c – R, T, d – P, T**	**a – P,R, b – P,Q, c – R, d – S**	**a – R,T, b – S,T, c – Q, d – P**
71	72	73	74	75	76	77	78	79	80
A – r, B – s, C – q, D – p	**AB**	**AD**	**B**	**D**	**CD**	**B**	**C**	**D**	**D**
81	82	83	84	85					
C	**C**	**B**	**A**	**D**					

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